

optimization section v2

kumarc

July 2026

1 Main Idea

We have a few interesting and novel parts of our paper. The majority of our methods are very comparable to existing methods in the political redistricting literature. The key difference between the school zoning and the political redistricting problem is that

1. the school district constraints are far more restrictive and even finding feasible solutions can be difficult, whereas for political redistricting it is more about proving optimality / generating a large number of solutions
2. we are able to fix centroids without excessive precomputation since we care less about optimality and schools serve as logical centroid locations.

Multi-level, MCMC, and MIP strategies have already been applied in the political redistricting literature. We offer a few key insights:

1. We apply political redistricting methods to school rezoning.
2. We concretely (albeit almost exclusively empirically) show that CP is better than MIP at this problem, with strong implications for the political redistricting literature.
3. We show that ReCom and relaxed ReCom can not effectively solve the school zoning problem (though short bursts and adaptive short bursts find significant success).
4. We provide a concrete multi-level strategy that is highly effective under tight constraints and allows us to have improved results; other multi-level strategies in the literature are likely to result in infeasible solutions.
5. We introduce a way to optimize for submodular choice set utility functions via logic-based Benders decomposition and show how it can be used as a heuristic for improving school choice outcomes.
6. We formulate market-clearing equilibrium (MID) and Sample Average Approximation (SAA) cutting-plane models to directly optimize utilitarian welfare over student budget sets.

7. We derive a congestion-priced upper bound on stable-matching welfare that is a function of the zoning rather than a constant, is valid for every non-negative price vector and every lottery seed, and — because it separates across applicants — admits an exact finite closed-form outer approximation requiring no recourse LP. This yields a cutting-plane method with finite convergence to a certified optimum of the bound, and prices the zone-confinement cost that the a priori welfare constants leave unpriced. Replicated over four seeds and two centroid configurations, it improves budget welfare over SAA by about 1% while stopping on its own separation certificate in roughly a tenth of the wall time.

Many of these ideas can (and arguably should be) used in the political redistricting literature.

2 Base Solvers

We represent a school district as an undirected, connected graph $G = (V, E)$, where each vertex $v \in V$ corresponds to a distinct geographic unit (e.g., a census block or block group) and each edge $e = \{u, v\} \in E$ indicates that two units are spatially adjacent. We seek to partition the graph into exactly k zones, denoted

$$\mathcal{P} = \{Z_1, Z_2, \dots, Z_k\}$$

which must satisfy:

1. $\bigcup_{i=1}^k Z_i = V$
2. $Z_i \cap Z_j = \emptyset \quad \forall i \neq j$
3. $Z_i \neq \emptyset \quad \forall i \in [k]$

We define the structural requirements of a valid school zoning partition $\mathcal{P}$ based on G , centroids C , number of zones k , and balance parameters δ .

2.1 Boundary Compactness (Objective Function)

To ensure that zones are geographically compact and pass the visual “eye test” for stakeholders, we evaluate the quality of a partition by its boundary size. The boundary of a partition $\mathcal{P}$, denoted $\partial\mathcal{P}$, is the set of all edges cut by the partition:

$$\partial\mathcal{P} = \{\{u, v\} \in E \mid u \in Z_i, v \in Z_j, i \neq j\}$$

Our fundamental objective is to find a partition that minimizes the total number of cut edges, $|\partial\mathcal{P}|$ (or, more generally, the total boundary length $\sum_{\{u,v\} \in \partial\mathcal{P}} w_{u,v}$ where $w_{u,v} \geq 0$ is the shared geographic boundary length).

2.2 Constraints

2.2.1 Centroid and Anchor Alignment

To reduce symmetry in our later formulations, each zone Z_i is anchored by a designated centroid vertex $c_i \in V$, where $C = \{c_1, c_2, \dots, c_k\}$. Schools serve as the natural geographic anchors for these centroids. A valid partition must structurally align each zone with exactly one unique centroid:

$$c_i \in Z_i \quad \forall i \in [k]$$

2.2.2 School Balance

Let S denote the total set of schools in the district. Also let $s(v) \in \mathbb{Z}_{\geq 0}$ denote the number of schools in a vertex $v \in V$. Note that this can potentially be greater than 1 for coarse graphs. Then we have $S(Z_i) = \sum_{v \in Z_i} s(v)$ for some zone $i \in [k]$. To prevent resource and choice imbalances, the total number of schools allocated to any given zone must be distributed as evenly as possible across the district:

$$\left\lfloor \frac{|S|}{k} \right\rfloor \leq S(Z_i) \leq \left\lceil \frac{|S|}{k} \right\rceil \quad \forall Z_i \in \mathcal{P}$$

2.2.3 Zone Contiguity

A zone cannot be split into geographically isolated pockets. Topologically, this requires that the induced subgraph of each zone must form a single connected component:

$$\text{The induced subgraph } G[Z_i] \text{ is connected } \forall Z_i \in \mathcal{P}$$

2.2.4 Capacity and Enrollment Tolerances

Each vertex $v \in V$ carries an associated estimated student population $n_v \in \mathbb{R}_{\geq 0}$ and a school seat capacity $c_v \in \mathbb{Z}_{\geq 0}$ (where $c_v > 0$ only if $v \in S$). For any zone Z_i , we define its total student enrollment as $N(Z_i) = \sum_{v \in Z_i} n_v$ and its total operational capacity as $C(Z_i) = \sum_{v \in Z_i} c_v$.

To protect schools from severe over-enrollment or under-utilization, the ratio of total seats to total students within any zone must fall within specified lower shortage tolerances (δ^-) and upper overage tolerances (δ^+):

$$(1 - \delta^-) \leq \frac{C(Z_i)}{N(Z_i)} \leq (1 + \delta^+) \quad \forall Z_i \in \mathcal{P}$$

2.2.5 Demographic Diversity and Socioeconomic Balance

Let $f_v \in \mathbb{R}_{\geq 0}$ denote the estimated count of students qualifying for Free and Reduced Lunch (FRL) programs at vertex v . The district-wide FRL proportion

is a fixed constant calculated as:

$$F = \frac{\sum_{v \in V} f_v}{\sum_{v \in V} n_v}$$

To counteract socioeconomic isolation, the localized FRL proportion of any individual zone Z_i must deviate from the global target F by no more than an allowable variance threshold $\delta_F \in \mathbb{R}_{>0}$:

$$\left| \frac{\sum_{v \in Z_i} f_v}{N(Z_i)} - F \right| \leq \delta_F \quad \forall Z_i \in \mathcal{P}$$

Let $\mathcal{I}$ represent the set of all valid input configurations for a problem with (G, C, k, δ) and $\mathcal{P}$ be the space of all possible partitions. Define the feasible set $\Omega(G, C, k, \delta) \subseteq \mathcal{P}$ as the subset of partitions that satisfy all constraints given the input (G, C, k, δ) .

We define the function $B : \mathcal{I} \rightarrow \mathcal{P} \cup \{\perp\}$ to map input configurations to a partition or the bottom element (representing infeasibility) as follows:

$$B(G, C, k, \delta) = \begin{cases} \arg \min_{\mathcal{P} \in \Omega(G, C, k, \delta)} |\partial \mathcal{P}| & \text{if } \Omega(G, C, k, \delta) \neq \emptyset \\ \perp & \text{if } \Omega(G, C, k, \delta) = \emptyset \end{cases} \quad (1)$$

Where $\perp$ denotes an infeasible input. Ties between solutions with identical cut edges are broken arbitrarily. All solvers will define various implementations of this base solver structure.

2.3 Math Programming

To solve this problem we will employ constraint programming (CP), a technique related to Mixed Integer Programming (MIP). CP differs from MIP in that it relies on boolean logic, domain reduction, and clause learning to efficiently prune search spaces, excelling in problems where combinatorial constraints impose severe barriers to finding feasible solutions. As we demonstrate empirically, CP-SAT significantly outperforms MIP backends on this problem, largely due to the combinatorial difficulty of satisfying the contiguity and balance constraints simultaneously. Notably, CP-SAT requires integer coefficients, so we employ integer scaling factors to encode constraints that involve fractional values.

We formulate the base model using binary assignment variables $x_{v,z} \in \{0, 1\}$, indicating whether geographic vertex $v \in V$ is assigned to zone $z \in [k]$, and cut boundary variables $b_{i,j} \in \{0, 1\}$ for each edge $(i, j) \in E$:

$$\min_{x,b} \sum_{(i,j) \in E} b_{i,j} \quad (2)$$

$$\text{s.t.} \quad \sum_{z \in [k]} x_{v,z} = 1 \quad \forall v \in V \quad (3)$$

$$x_{c(z),z} = 1 \quad \forall z \in [k] \quad (4)$$

$$x_{v,z} = 1 \implies \bigvee_{u \in \text{cn}(v, c(z))} x_{u,z} \quad \forall z \in [k], v \in V \setminus \{c(z)\} \quad (5)$$

$$\left\lfloor \frac{|S|}{k} \right\rfloor \leq \sum_{v \in V} s(v) x_{v,z} \leq \left\lceil \frac{|S|}{k} \right\rceil \quad \forall z \in [k] \quad (6)$$

$$(1 - \delta^-)N(Z_z) \leq C(Z_z) \leq (1 + \delta^+)N(Z_z) \quad \forall z \in [k] \quad (7)$$

$$(F - \delta_F)N(Z_z) \leq \sum_{v \in V} f_v x_{v,z} \leq (F + \delta_F)N(Z_z) \quad \forall z \in [k] \quad (8)$$

$$(x_{i,z} \wedge \neg x_{j,z}) \vee (x_{j,z} \wedge \neg x_{i,z}) \implies b_{i,j} = 1 \quad \forall (i,j) \in E, z \in [k] \quad (9)$$

$$x_{v,z} \in \{0, 1\} \quad \forall v \in V, z \in [k] \quad (10)$$

$$b_{i,j} \in \{0, 1\} \quad \forall (i,j) \in E \quad (11)$$

Each inequality directly realizes one of the structural requirements introduced in Section 2.2:

- **Objective (2):** Minimizes the total number of cut edges across all zone boundaries.
- **Unique Assignment (3):** Enforces that every geographic unit $v \in V$ is assigned to exactly one zone $z \in [k]$.
- **Centroid Anchoring (4):** Fixes each designated zone centroid $c(z)$ to zone z , eliminating zone permutation symmetries.
- **Zone Contiguity (5):** Enforces graph connectedness through a rooted closer-neighbor condition based on the formulation of Shirabe (2005) and Validi et al. (2022). For each vertex $v \neq c(z)$ and zone z , let $\text{cn}(v, c(z))$ denote the set of adjacent neighbors that are strictly closer to centroid $c(z)$ in spatial geometry:

$$\text{cn}(v, c(z)) = \{u \in N(v) \mid \text{dist}(u, c(z)) < \text{dist}(v, c(z))\}$$

Constraint (5) stipulates that if vertex v joins zone z , at least one neighbor $u \in \text{cn}(v, c(z))$ must also join zone z . Because distances to $c(z)$ strictly decrease along this chain, this constraint induces a cycle-free directed path from every assigned vertex back to the centroid $c(z)$, ensuring that the induced subgraph $G[Z_z]$ is connected.

- **School Count Balance (6):** Distributes the $|S|$ schools evenly among all k zones, restricting zone z 's school count between $\lfloor |S|/k \rfloor$ and $\lceil |S|/k \rceil$.
- **Capacity Tolerances (7):** Ensures that the aggregate seat capacity within zone z remains within lower shortage tolerance δ^- and upper overage tolerance δ^+ of the resident student population.
- **Socioeconomic Balance (8):** Bounds zone z 's Free and Reduced Lunch (FRL) proportion within a maximum deviation δ_F from the district-wide mean F . (The corresponding multi-group racial and ethnic diversity balance constraints follow an analogous form and are detailed in the Appendix).
- **Cut Edge Implication (9):** Enforces that $b_{i,j} = 1$ via Boolean half-reification whenever endpoints i and j of edge $(i,j) \in E$ are assigned to different zones (i.e., if vertex i joins zone z but j does not, or vice versa).

Equivalently, CP-SAT receives these constraints in zero-right-hand-side form. Because CP-SAT requires integer coefficients, each inequality is scaled by a large integer multiplier M (typically $M = 10^4$) and its coefficients are rounded conservatively before being added to the model. For example, the lower FRL constraint is written as:

$$\sum_{v \in V} \lfloor M \cdot (f_v - (F - \delta_F)n_v) \rfloor \cdot x_{v,z} \geq 0 \quad \forall z \in [k]$$

where the floor function preserves conservative bounding without weakening the feasibility guarantees.

2.4 MCMC Models

In the political redistricting literature, Markov Chain Monte Carlo (MCMC) methods—most prominently Recombination (ReCom)—are widely used to sample vast ensembles of contiguous districts. In political redistricting, the primary constraint is typically equal population balance (a single scalar target). In contrast, school rezoning introduces a substantially more restrictive constraint manifold defined by capacity bounds, tight demographic diversity bounds, and exact discrete school count allocations.

2.4.1 ReCom Proposal Kernel

All ReCom-family variants share a common combinatorial transition kernel operating on partition $\mathcal{P}_t = \{Z_1, \dots, Z_k\}$ of graph $G = (V, E)$:

1. **Adjacent Zone Pair Selection:** Let $\mathcal{A}(\mathcal{P}_t)$ denote the set of pairs of zones sharing at least one boundary edge:

$$\mathcal{A}(\mathcal{P}_t) = \{(a, b) \mid 1 \leq a < b \leq k, \exists (u, v) \in E \text{ with } u \in Z_a, v \in Z_b\} \quad (12)$$

A pair (a, b) is drawn uniformly at random from $\mathcal{A}(\mathcal{P}_t)$.

2. **Uniform Spanning Tree Sampling:** Form the merged region $U = Z_a \cup Z_b$. Because Z_a and Z_b are individually contiguous and share boundary edges, the induced subgraph $G[U]$ is connected. A spanning tree $T = (U, E(T))$ is sampled uniformly from the set $\mathcal{T}(G[U])$ of all spanning trees of $G[U]$ via Wilson’s loop-erased random walk algorithm (Wilson, 1996), ensuring that each tree is drawn with exact probability $1/|\mathcal{T}(G[U])|$.
3. **Spanning Tree Cuts and Zone Contiguity:** Tree T is rooted at an arbitrary root vertex $r \in U$. Each non-root vertex $u \in U \setminus \{r\}$ and its directed edge to its parent $(u, \text{parent}(u)) \in E(T)$ uniquely defines a tree bisection into subtree T_u and complement $U \setminus T_u$. Because T spans $G[U]$, the induced subgraphs $G[T_u]$ and $G[U \setminus T_u]$ are connected by construction, guaranteeing topological contiguity without post-hoc connectivity checks.
4. **Candidate Reassignments:** Each subtree cut yields two candidate assignments:

$$\text{Orientation 1: } Z'_a = T_u, \quad Z'_b = U \setminus T_u \quad (13)$$

$$\text{Orientation 2: } Z'_a = U \setminus T_u, \quad Z'_b = T_u \quad (14)$$

Orientations that violate designated centroid anchoring ($c_a \notin Z'_a$ or $c_b \notin Z'_b$) or explicit vertex candidate restrictions are pruned, producing a legal candidate cut pool $\mathcal{C}$.

5. **Fast Additive Evaluation:** Vertex attributes (students n_v , capacity c_v , schools $s(v)$, and demographic counts f_v) are additive set functions. Subtree statistics for each candidate cut $c \in \mathcal{C}$ are evaluated in a single post-order traversal over T , e.g., $N(T_u) = n_u + \sum_{w \in \text{children}(u)} N(T_w)$ and $N(U \setminus T_u) = N(U) - N(T_u)$. The updated boundary cost is computed from non-tree edge crossings via lowest common ancestor prefix sums:

$$|\partial \mathcal{P}_c| = |\partial \mathcal{P}_t| - |\partial(Z_a, Z_b)| + |\partial(Z_a^{(c)}, Z_b^{(c)})| \quad (15)$$

2.4.2 Standard ReCom

Standard ReCom (DeFord et al., 2021) evaluates candidate proposals under a hard feasibility filter:

1. **Initial Seeding:** Because no valid baseline map exists for school re-zoning, the chain is initialized via a Voronoi diagram anchored at designated school centroids $C = \{c_1, \dots, c_k\}$, where each vertex v is assigned to its geographically nearest centroid: $z^{(0)}(v) = \arg \min_{z \in [k]} \text{dist}(v, c_z)$. Disconnected components are reallocated to adjacent zones maintaining centroid paths, producing an initial contiguous partition $\mathcal{P}_0$. In general, $\mathcal{P}_0 \notin \Omega(G, C, k, \delta)$.
2. **Proposal and Rejection:** Let $\mathcal{C}_{\text{feas}} = \{c \in \mathcal{C} \mid \mathcal{P}_c \in \Omega(G, C, k, \delta)\}$ be the set of candidate cuts where both prospective zones satisfy all capacity,

school count, and demographic constraints. If $\mathcal{C}_{\text{feas}} \neq \emptyset$, a candidate cut is drawn uniformly: $c \sim \text{Uniform}(\mathcal{C}_{\text{feas}})$; otherwise, $c \sim \text{Uniform}(\mathcal{C})$. The transition is accepted if and only if the prospective partition is globally feasible:

$$\mathcal{P}_{t+1} = \begin{cases} \mathcal{P}_c & \text{if } \mathcal{P}_c \in \Omega(G, C, k, \delta) \\ \mathcal{P}_t & \text{if } \mathcal{P}_c \notin \Omega(G, C, k, \delta) \end{cases} \quad (16)$$

Theoretical Bottleneck on School Rezoning: In political redistricting, baseline legislative plans start within the feasible manifold Ω , and the only primary constraint is equal population. In school rezoning, Ω is the simultaneous intersection of $m \geq 6k$ discrete constraints (exact school counts $\lfloor |S|/k \rfloor \leq S(Z) \leq \lceil |S|/k \rceil$, tight capacity bounds, and FRL balance). Because constraints are enforced post-cut, the probability that a uniform spanning tree cut simultaneously satisfies all bounds is vanishingly small: $P(c \in \mathcal{C}_{\text{feas}}) \approx 0$. As a result, standard ReCom rejects virtually all moves and remains trapped at $\mathcal{P}_0$.

2.4.3 Relaxed ReCom

To navigate through infeasible space and bypass zero acceptance, Relaxed ReCom eliminates the hard rejection step ($P_{\text{accept}} = 1$) and samples candidate cuts from a biased categorical distribution that steers the chain toward feasibility and geometric compactness.

For each candidate cut $c \in \mathcal{C}$ producing prospective zones $Z_a^{(c)}$ and $Z_b^{(c)}$, we compute the unnormalized scoring function $\mu(c)$:

$$\ln \mu(c) = \theta_{\text{trees}} \left(\text{cr}(Z_a^{(c)}) + \text{cr}(Z_b^{(c)}) \right) + \sum_{\kappa \in \mathcal{K}} \theta_{\kappa} \left(\ln \kappa(Z_a^{(c)}) + \ln \kappa(Z_b^{(c)}) \right) \quad (17)$$

where:

- $\text{cr}(Z) = \max(0, |E[Z]| - |V(Z)| + 1)$ is the cycle rank (cyclomatic number) of the induced subgraph $G[Z]$. By Kirchhoff’s Matrix Tree Theorem, the number of spanning trees $|\mathcal{T}(G[Z])|$ scales exponentially with the cycle rank: $|\mathcal{T}(G[Z])| \approx \exp(\text{cr}(Z))$. Weighting by $\theta_{\text{trees}} = 1$ penalizes stringy, bottlenecked boundaries and favors internally dense, compact subgraphs.
- $\mathcal{K}$ is the set of balance metrics evaluated on each prospective zone:
 1. Node count: $\kappa_{\text{nodes}}(Z) = |V(Z)|$, with weight $\theta_{\text{nodes}} = 1$
 2. Student enrollment: $\kappa_{\text{students}}(Z) = N(Z)$, with weight $\theta_{\text{students}} = 1$
 3. Seat capacity: $\kappa_{\text{seats}}(Z) = C(Z)$, with weight $\theta_{\text{seats}} = 1$
 4. Socioeconomic need: $\kappa_{\text{frl}}(Z) = F(Z)$, with weight $\theta_{\text{frl}} = 3$
 5. School count: $\kappa_{\text{sch}}(Z) = S(Z)$, with large weight $\theta_{\text{sch}} = 45$ to heavily penalize school count disparities
 6. Relative capacity mismatch: $\kappa_{\text{shortage\%}}(Z) = \max\left(\frac{|N(Z) - C(Z)|}{\max(N(Z), \epsilon)}, \epsilon\right)$ with weight $\theta_{\text{shortage\%}} = 10$ and $\epsilon = 10^{-12}$.

Candidate cut c is drawn from the categorical distribution:

$$P(c) = \frac{\mu(c)}{\sum_{c' \in \mathcal{C}} \mu(c')} \quad (18)$$

The selected move is accepted unconditionally ($\mathcal{P}_{t+1} \leftarrow \mathcal{P}_c$). While Relaxed ReCom traverses infeasible states freely, unconstrained random walks with static weights drift over long trajectories and frequently fail to settle into the narrow intersection of all feasibility constraints.

2.4.4 Short Bursts

Building upon the Short Bursts framework (Cannon et al., 2020), we replace the unbounded random walk with repeated localized search bursts of fixed length L (typically $L = 25$). Starting from an incumbent partition $\mathcal{P}^{(b)}$ at burst iteration b , the chain executes L unconstrained transitions using the ReCom or Relaxed ReCom proposal kernel, generating a trajectory of visited states $\mathcal{S}^{(b)} = \{\mathcal{P}_1, \dots, \mathcal{P}_L\}$.

To compare intermediate partitions across heterogeneous constraints with disparate numerical scales (e.g., student counts $\sim 10^3$, school counts $\sim 10^0$, demographic proportions $\sim 10^{-1}$), we track the running maximum observed violation for each constraint $j \in [m]$ across all visited states up to step τ :

$$M_j^{(\tau)} = \max_{0 \leq t \leq \tau} v_j(\mathcal{P}_t) \quad (19)$$

where $v_j(\mathcal{P}) = \sum_{z=1}^k v_j(Z_z) \geq 0$ is the total linear violation slack for constraint j . The dynamically normalized violation penalty is defined as:

$$\Phi(\mathcal{P}) = \sum_{j: M_j^{(\tau)} > 0} \frac{v_j(\mathcal{P})}{M_j^{(\tau)}} \quad (20)$$

Partitions are evaluated under a strict lexicographic Pareto ordering $\prec$ that prioritizes feasibility before optimizing boundary compactness:

$$\mathcal{P} \prec \mathcal{Q} \iff \begin{cases} \mathcal{P} \in \Omega \quad \text{and} \quad \mathcal{Q} \notin \Omega \\ \mathcal{P}, \mathcal{Q} \in \Omega \quad \text{and} \quad |\partial \mathcal{P}| < |\partial \mathcal{Q}| \\ \mathcal{P}, \mathcal{Q} \notin \Omega \quad \text{and} \quad \Phi(\mathcal{P}) < \Phi(\mathcal{Q}) - \epsilon_{\text{tol}} \end{cases} \quad (21)$$

with tolerance $\epsilon_{\text{tol}} = 10^{-6}$. At the conclusion of burst b , the best visited state is identified:

$$\mathcal{P}^* = \arg \min_{\mathcal{P} \in \mathcal{S}^{(b)}} \mathcal{P} \quad \text{under } \prec \quad (22)$$

If $\mathcal{P}^* \prec \mathcal{P}^{(b)}$, the improvement is accepted as the new incumbent: $\mathcal{P}^{(b+1)} \leftarrow \mathcal{P}^*$; otherwise, the chain resets to the incumbent: $\mathcal{P}^{(b+1)} \leftarrow \mathcal{P}^{(b)}$. This ratcheting mechanism allows the chain to explore infeasible configurations locally while guaranteeing that the search never drifts permanently into worse regions.

2.4.5 Adaptive Short Bursts

Although Short Bursts guides the search restarts, proposal generation within each burst remains unguided. When certain constraints (such as school count balance) are far more rigid than others, random tree bisections rarely decrease multiple violations simultaneously. We address this by formulating Adaptive Short Bursts, which dynamically couples an augmented Lagrangian proposal distribution with Adam dual multiplier ascent.

Percentage-Normalized Constraint Violations: Constraint residuals are converted into fractional percentage deviations:

$$\tilde{v}_{\text{cap}}(Z) = \frac{v_{\text{cap}}(Z)}{N(Z)} \times 100\%, \quad \tilde{v}_{\text{sch}}(Z) = \frac{v_{\text{sch}}(Z)}{\bar{S}} \times 100\% \quad (23)$$

where $\bar{S} = |S|/k$ is the target average school count per zone.

Augmented Lagrangian Energy Formulation: Let $\boldsymbol{\eta} = (\eta_1, \dots, \eta_m) \in \mathbb{R}_+^m$ denote dual penalty multipliers, initialized to $\eta_j^{(0)} = 1.0$. For each candidate cut $c \in \mathcal{C}$ yielding prospective partition $\mathcal{P}_c$, we define the candidate energy function:

$$E(c; \mathbf{w}) = |\partial \mathcal{P}_c| + \sum_{j=1}^m w_j \cdot \tilde{v}_j(\mathcal{P}_c)^2 \quad (24)$$

where penalty weights w_j are dynamically coupled to the incumbent partition's cut boundary cost:

$$w_j = \max \left(1.0, |\partial \mathcal{P}^{(b)}| + \eta_j \cdot \tilde{v}_j(\mathcal{P}^{(b)})^2 \right) \quad (25)$$

Softmax Proposal Distribution: Within each burst, candidate cuts are sampled from a Boltzmann distribution parameterized by temperature $T > 0$ (default $T = 1.0$):

$$P(c) = \frac{\exp \left(-\frac{E(c; \mathbf{w})}{T} \right)}{\sum_{c' \in \mathcal{C}} \exp \left(-\frac{E(c'; \mathbf{w})}{T} \right)} \quad (26)$$

This exponentially suppresses candidate cuts that worsen heavily penalized violations, actively steering spanning tree bisections toward the feasible boundary.

Adam Dual Ascent Across Bursts: At the end of burst b , after selecting the best partition $\mathcal{P}^*$ via the lexicographic ordering $\prec$, the dual ascent subgradient is computed from the squared violations of $\mathcal{P}^*$:

$$g_j^{(b)} = \tilde{v}_j(\mathcal{P}^*)^2 \quad \forall j \in [m] \quad (27)$$

The dual multipliers $\boldsymbol{\eta}$ are updated using projected Adam optimization:

$$\mathbf{m}_b = \beta_1 \mathbf{m}_{b-1} + (1 - \beta_1) \mathbf{g}^{(b)} \quad (28)$$

$$\mathbf{u}_b = \beta_2 \mathbf{u}_{b-1} + (1 - \beta_2) (\mathbf{g}^{(b)} \odot \mathbf{g}^{(b)}) \quad (29)$$

$$\hat{\mathbf{m}}_b = \frac{\mathbf{m}_b}{1 - \beta_1^b}, \quad \hat{\mathbf{u}}_b = \frac{\mathbf{u}_b}{1 - \beta_2^b} \quad (30)$$

$$\eta_j^{(b+1)} = \max \left(0.0, \eta_j^{(b)} + \gamma \frac{\hat{m}_{j,b}}{\sqrt{\hat{u}_{j,b} + \epsilon_{\text{Adam}}}} \right) \quad (31)$$

with learning rate $\gamma = 0.1$, moment decay rates $\beta_1 = 0.9, \beta_2 = 0.999$, and $\epsilon_{\text{Adam}} = 10^{-8}$. The updated multipliers $\boldsymbol{\eta}^{(b+1)}$ and incumbent $\mathcal{P}^{(b+1)}$ then determine the proposal weights $\mathbf{w}$ for the subsequent burst.

Closed-Loop Feedback Dynamics: If a constraint j remains persistently violated across bursts ($\tilde{v}_j > 0$), its gradient $g_j > 0$ inflates η_j and w_j , exponentially suppressing candidate cuts that fail to satisfy constraint j . When constraint j is satisfied ($\tilde{v}_j = 0$), $g_j = 0$, allowing the second moment to decay and stabilizing η_j . Once all constraints reach feasibility ($\tilde{\mathbf{v}} = \mathbf{0}$), the energy collapses to $E(c; \mathbf{w}) = |\partial \mathcal{P}_c|$, and the Boltzmann distribution purely drives cut boundary minimization.

3 Hierarchical Graph Contraction

The computational complexity of the zoning problem scales rapidly with the number of vertices $|V|$ and zones k . Informally, the discrete search space contains $k^{|V|}$ possible assignments. Solving this directly at the census block level (thousands of nodes) is often computationally intractable. To resolve this, we develop a principled graph contraction hierarchy that creates arbitrary-fidelity coarsened graphs.

Let our base graph be $G_0 = (V_0, E_0)$. We construct a contracted graph $G_1 = (V_1, E_1)$ with $|V_1| = k_1 < |V_0|$ vertices that preserves the topological and spatial structure of G_0 . Formally, G_1 is a valid contraction of G_0 under a surjective mapping $\phi_1 : V_0 \rightarrow V_1$ if:

Connectivity: For every vertex $u \in V_1$, the induced preimage subgraph $G_0[\phi_1^{-1}(u)]$ is connected.

Adjacency: For any two distinct vertices $u, v \in V_1$, an edge exists in G_1 if and only if their preimages are adjacent in G_0 :

$$(u, v) \in E_1 \iff \exists x \in \phi_1^{-1}(u), y \in \phi_1^{-1}(v) \text{ such that } (x, y) \in E_0$$

We compute this partitioning using KaHIP (Sanders and Schulz, 2013). Crucially, we do not contract school vertices. Let $S \subset V_0$ denote the set of vertices containing schools. When constructing any coarse graph $G_{\ell+1}$ from

G_ℓ , every school node $s \in S$ is held out as an indivisible singleton vertex ($V_{\ell+1}^{\text{school}} = S$). KaHIP partitions only the remaining non-school geographic units $V_\ell \setminus S$ into $k_{\ell+1} - |S|$ contiguous blocks. Coarse edges between aggregated residential blocks and school singletons are derived from crossing base edges: $(u, s) \in E_{\ell+1} \iff \exists x \in \phi^{-1}(u)$ such that $(x, s) \in E_\ell$.

Preserving school nodes as singletons is essential for three reasons:

1. **School count exactness:** It ensures $s(v) = 1$ for each school node and $s(v) = 0$ for all non-school aggregates, enabling exact enforcement of the school balance constraint $\lfloor |S|/k \rfloor \leq S(Z_i) \leq \lceil |S|/k \rceil$ at any graph hierarchy level. If schools were merged into aggregate blocks, multiple schools could be bundled into a single coarse vertex, potentially rendering the school balance constraint infeasible.
2. **Centroid anchor preservation:** Distinct school singletons ensure that designated zone centroids $c_i \in S$ remain individually assignable and are never co-contracted into the same coarse vertex.
3. **Capacity localization:** School seat capacities c_s remain precisely associated with their geographic points, preserving spatial accuracy in capacity-to-student balance.

4 Strategies

4.1 Multilevel Strategy

While coarsened graphs can be solved independently, their primary utility lies in a multi-level optimization strategy. Specifically, we solve the zoning problem hierarchically across a sequence of levels $G_n, G_{n-1}, \dots, G_0$, from coarsest to finest.

At each transition from level i to level $i - 1$, the coarser partition $\mathcal{P}_i$ is projected onto the finer graph G_{i-1} using the contraction mapping ϕ_i . Vertices situated deeply within the interior of a projected zone—whose graph distance to the zone boundary exceeds a specified radius r —have their assignments fixed to that zone. Only vertices near the boundary (within distance r) are treated as candidate decision variables in the finer optimization solve. The projected partition also provides a high-quality warm-start hint. This restricts the decision space at fine levels exclusively to boundary refinement, yielding substantial computational speedups.

This approach yields two major practical advantages:

1. We find higher-quality, more compact solutions within the same time limit compared to solving fine graphs from scratch.
2. We reliably discover feasible solutions at the census block level in parameter regimes where monolithic fine-level solvers fail completely.

Algorithm 1 Multilevel Strategy

Require: Sequence of graphs $G_n, G_{n-1}, \dots, G_0$ from coarsest to finest, base solver B , parameters δ , boundary relaxation radius r .

```
1: procedure MULTILEVEL( $\{G_i\}_{i=n}^0, B, \delta, r$ )
2:    $\mathcal{P}_n \leftarrow B(G_n, C, k, \delta)$  ▷ Solve coarsest level globally
3:   for  $i \leftarrow n - 1$  down to 0 do
4:      $\tilde{\mathcal{P}}_i \leftarrow \text{Project}(\mathcal{P}_{i+1}, G_i)$  ▷ Project partition via contraction map
5:      $V^{\text{bound}} \leftarrow \{u \in V_i \mid \exists v \in N(u) \text{ s.t. } \tilde{\mathcal{P}}_i(u) \neq \tilde{\mathcal{P}}_i(v)\}$ 
6:      $\mathcal{B}_r \leftarrow \{v \in V_i \mid \text{dist}_{G_i}(v, V^{\text{bound}}) \leq r\}$  ▷ Boundary neighborhood
7:     Set candidate zones  $\mathcal{C}(v) \leftarrow \{\tilde{\mathcal{P}}_i(v)\}$  for all  $v \in V_i \setminus \mathcal{B}_r$  ▷ Fix interior
8:     Set candidate zones  $\mathcal{C}(v) \leftarrow [k]$  for all  $v \in \mathcal{B}_r$  ▷ Relax boundary
9:      $\mathcal{P}_i \leftarrow B(G_i, C, k, \delta \mid \text{hint} = \tilde{\mathcal{P}}_i, \text{candidates} = \mathcal{C})$ 
10:   end for
11:   return  $\mathcal{P}_0$ 
12: end procedure
```

5 Choice Set Optimization

Instead of solely evaluating zoning by geometric compactness, we now consider the academic welfare that a zone configuration delivers to families. Under a zoned choice system, a family residing in vertex v gains access to the set of schools assigned to that zone, $S' \subseteq S$. We wish to optimize the expected utility that families derive from their available choice set.

Under a standard Multinomial Logit (MNL) discrete choice model (McFadden, 1974), each student derives utility $U_{i,s} = V_s + \varepsilon_{i,s}$ from school s , where V_s is the deterministic mean utility and $\varepsilon_{i,s} \stackrel{\text{i.i.d.}}{\sim} \text{Gumbel}(0, 1)$ represents an idiosyncratic taste shock following a standard Type I Extreme Value distribution. The expected maximum utility of choice set S' is given by the log-sum-exp formula:

$$U(S') = \mathbb{E} \left[\max_{s \in S'} (V_s + \varepsilon_{i,s}) \right] = \ln \left(\sum_{s \in S'} e^{V_s} \right) + \gamma \quad (32)$$

where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant.

5.1 Gain Bound

Because the log-sum-exp function is submodular, the marginal gain from adding an additional school $s \in S \setminus S'$ exhibits diminishing marginal returns: for any $S' \subset T \subseteq S$,

$$\Delta_{s,S'}^+ = U(S' \cup \{s\}) - U(S') \geq U(T \cup \{s\}) - U(T) \quad (33)$$

We sum these bounds to establish an upper bound on the gain from adding an arbitrary subset of schools $G = \{s_1, \dots, s_{n_{\text{gain}}}\} \subseteq S \setminus S'$. Construct the intermediate sequence $S_0 = S'$ and $S_j = S_{j-1} \cup \{s_j\}$ for $j = 1, \dots, n_{\text{gain}}$,

terminating at $S_{n_{\text{gain}}} = S' \cup G$. Using a telescoping sum and submodularity ($S' \subseteq S_{j-1}$), we have:

$$\begin{aligned} U(S' \cup G) - U(S') &= \sum_{j=1}^{n_{\text{gain}}} (U(S_j) - U(S_{j-1})) \\ &\leq \sum_{j=1}^{n_{\text{gain}}} (U(S' \cup \{s_j\}) - U(S')) = \sum_{s \in G} \Delta_{s, S'}^+ \end{aligned} \quad (34)$$

5.2 Loss Bound

Similarly, we establish a lower bound on the utility lost from removing a school $s \in S'$. Submodularity implies that the loss incurred by removing s from the complete district set S is smaller than the loss from removing it from any smaller subset $S' \subseteq S$:

$$\Delta_s^- = U(S) - U(S \setminus \{s\}) \leq U(S') - U(S' \setminus \{s\}) \quad (35)$$

Crucially, Δ_s^- depends only on the global set S and provides a uniform lower bound for any choice set S' .

For an arbitrary subset of lost schools $L = \{s_1, \dots, s_{n_{\text{loss}}}\} \subseteq S'$, construct the sequence $S_0 = S'$ and $S_j = S_{j-1} \setminus \{s_j\}$ ending at $S_{n_{\text{loss}}} = S' \setminus L$. Telescoping gives:

$$\begin{aligned} U(S') - U(S' \setminus L) &= \sum_{j=1}^{n_{\text{loss}}} (U(S_{j-1}) - U(S_j)) \geq \sum_{s \in L} \Delta_s^- \\ U(S' \setminus L) &\leq U(S') - \sum_{s \in L} \Delta_s^- \end{aligned} \quad (36)$$

5.3 Combined Upper Bound

Combining the gain and loss bounds, we obtain an outer linear approximation for the utility of any arbitrary alternative choice set $\tilde{S} \subseteq S$ evaluated from baseline S' . Let $G = \tilde{S} \setminus S'$ be the gained schools and $L = S' \setminus \tilde{S}$ be the lost schools:

$$U(\tilde{S}) \leq U(S') + \sum_{s \in \tilde{S} \setminus S'} \Delta_{s, S'}^+ - \sum_{s \in S' \setminus \tilde{S}} \Delta_s^- \quad (37)$$

5.4 Logic-Based Benders Decomposition

To maximize total student utility while maintaining compactness, we reformulate cut edge minimization as a budget constraint:

$$\sum_{(i,j) \in E} b_{i,j} \leq \delta_{\text{cut}} |E|$$

where $\delta_{\text{cut}} \in (0, 1)$ sets the maximum allowable cut edge proportion.

Let $a_{v,l(s)} \in \{0,1\}$ indicate whether vertex v and school location $l(s)$ are assigned to the same zone:

$$a_{v,l(s)} \iff \bigvee_{z \in [k]} (x_{v,z} \wedge x_{l(s),z}) \quad (38)$$

which is linearized via McCormick envelopes in MIP or boolean operators in CP-SAT.

We introduce auxiliary decision variables $\hat{U}_v \in \mathbb{R}$ representing the total expected choice set utility of residents at vertex $v \in V$, and solve the resulting problem via logic-based Benders decomposition:

Master Problem: Maximize $\sum_{v \in V} \hat{U}_v$ subject to the base structural constraints (3)–(8), the compactness budget, access definitions (38), and accumulated Benders cuts. In iteration 0, $\hat{U}_v$ is unbounded from above, yielding an initial feasible partition $\mathcal{P}^0 = (x^0, z^0)$.

Subproblem: At iteration t , given zoning partition z^t , evaluate the exact choice set available at each vertex v , $S_v^t = \{s \in S \mid z^t(l(s)) = z^t(v)\}$, and compute true utility $U_v(S_v^t)$ alongside marginal gains $\Delta_{t,s,v}^+ = U_v(S_v^t \cup \{s\}) - U_v(S_v^t)$ and global losses $\Delta_{s,v}^- = U_v(S) - U_v(S \setminus \{s\})$.

Benders Optimality Cuts: Add unconditional linear outer approximation cuts to the master problem:

$$\hat{U}_v \leq U_v(S_v^t) + \sum_{s \in S \setminus S_v^t} \Delta_{t,s,v}^+ a_{v,l(s)} - \sum_{s \in S_v^t} \Delta_{s,v}^- (1 - a_{v,l(s)}) \quad \forall v \in V \quad (39)$$

Alternatively, cuts may be aggregated across all vertices into a single total utility bound:

$$\sum_{v \in V} \hat{U}_v \leq \sum_{v \in V} \left(U_v(S_v^t) + \sum_{s \in S \setminus S_v^t} \Delta_{t,s,v}^+ a_{v,l(s)} - \sum_{s \in S_v^t} \Delta_{s,v}^- (1 - a_{v,l(s)}) \right) \quad (40)$$

Theorem 1 (Finite Convergence and Optimality). *The logic-based Benders decomposition terminates in a finite number of iterations at a globally optimal zone partition $\mathcal{P}^*$.*

Proof. Let $\mathcal{X}$ denote the finite set of feasible binary zone assignment vectors $x \in \{0,1\}^{|V| \times k}$ satisfying the base constraints. First, by inequality (37), every generated Benders cut (39) is a valid global upper bound: for any $x \in \mathcal{X}$, the master variable satisfies $\hat{U}_v(x) \geq U_v(S_v(x))$ for all v . Second, the cut is exact at the evaluated point x^t : setting $x = x^t$, we have $a_{v,l(s)} = 0$ for all $s \notin S_v^t$ and $(1 - a_{v,l(s)}) = 0$ for all $s \in S_v^t$, which collapses the right-hand side of (39) exactly to $U_v(S_v^t)$. Thus $\hat{U}_v(x^t) = U_v(S_v^t)$. Finally, because $\mathcal{X}$ is finite, the master problem cannot generate infinitely many distinct solutions. When the

master problem produces an incumbent partition x^{t+1} whose objective matches its true subproblem evaluation, we have:

$$\sum_{v \in V} U_v(S_v(x^{t+1})) = \sum_{v \in V} \hat{U}_v(x^{t+1}) \geq \sum_{v \in V} \hat{U}_v(x) \geq \sum_{v \in V} U_v(S_v(x)) \quad \forall x \in \mathcal{X}$$

proving that x^{t+1} is a global maximum. ■

6 Budget Set Optimization

Choice set optimization maximizes the expected utility of the options accessible to a family. However, in over-subscribed urban school districts, access does not guarantee enrollment: admission is governed by centralized market clearing under school priority tiers and lottery tiebreakers. We now formulate budget set optimization to maximize total utilitarian welfare under actual admissions.

6.1 Market-Clearing Cutoff (MID) Optimization

We represent student demand through compressed applicant types $g \in \mathcal{G}$, where n_g is the number of students of type g . Each type g is characterized by a resident node $v(g)$, an ordered preference list of programs $(s(g, 1), s(g, 2), \dots, s(g, R_g))$, corresponding priority tiers $\rho(g, r) \in \mathbb{Z}_{\geq 0}$, and utilities $u(g, r)$ satisfying $u(g, 1) \geq u(g, 2) \geq \dots \geq u(g, R_g) > 0$.

Each program $s \in \mathcal{S}$ has capacity q_s and is assigned a continuous market-clearing cutoff $P_s \in \mathbb{R}_{\geq 0}$. Let $\omega \in [0, 1]$ denote an applicant's single uniform lottery number. An applicant in priority tier ρ qualifies for program s if and only if $\rho + \omega \geq P_s$, or equivalently $\omega \geq P_s - \rho$. The rejected lottery mass is therefore:

$$t_{s, \rho} = \min(1, \max(P_s - \rho, 0)) \quad (41)$$

For choice r of type g , access depends on the zoning assignment x . Let $a_{g, s} \in \{0, 1\}$ indicate whether program s is accessible to type g ($a_{g, s} = 1$ if citywide or located in type g 's assigned zone). The effective rejected mass is:

$$e_{g, r} = \begin{cases} t_{s(g, r), \rho(g, r)} & \text{if } a_{g, s(g, r)} = 1 \\ 1 & \text{if } a_{g, s(g, r)} = 0 \end{cases} \quad (42)$$

An applicant enrolls at choice r if and only if they are rejected from all preferred choices $1, \dots, r-1$ and qualify at choice r . Let $R_{g, r}$ denote the remaining lottery mass rejected by choices $1, \dots, r$. Assignment follows the recurrence relation:

$$R_{g, 0} = 1, \quad R_{g, r} = \min(R_{g, r-1}, e_{g, r}) \quad \forall r \in [R_g] \quad (43)$$

The assigned student mass at choice r is then:

$$d_{g, r} = R_{g, r-1} - R_{g, r} = \max(0, R_{g, r-1} - e_{g, r}) \quad (44)$$

Proposition 1 (Recurrence Correctness). *The mass recurrence (43) exactly matches the lottery admission interval.*

Proof. Under uniform lottery tiebreaker $\omega \in [0, 1]$, applicant g is rejected by alternative k if and only if $\omega < e_{g,k}$. Consequently, g is rejected by all top r alternatives if and only if $\omega < \min_{k=1}^r e_{g,k}$. Defining $R_{g,r} = \min_{k=1}^r e_{g,k}$, we have $R_{g,0} = 1$ and:

$$R_{g,r} = \min_{k=1}^r e_{g,k} = \min \left(\min_{k=1}^{r-1} e_{g,k}, e_{g,r} \right) = \min(R_{g,r-1}, e_{g,r})$$

The measure of the set of lottery numbers where applicant g is rejected by choices $1, \dots, r-1$ ($\omega < R_{g,r-1}$) but accepted at choice r ($\omega \geq e_{g,r}$) is the interval length $[\min(R_{g,r-1}, e_{g,r}), R_{g,r-1}]$, which is precisely $R_{g,r-1} - \min(R_{g,r-1}, e_{g,r}) = R_{g,r-1} - R_{g,r}$. ■

Total program demand cannot exceed seat capacity:

$$D_s(x, P) = \sum_{g \in \mathcal{G}} \sum_{r: s(g,r)=s} n_g d_{g,r} \leq q_s \quad \forall s \in \mathcal{S} \quad (45)$$

The MID optimization problem maximizes total utilitarian welfare:

$$\max_{x \in \Omega, P \geq 0} \sum_{g \in \mathcal{G}} \sum_{r=1}^{R_g} n_g u(g, r) d_{g,r} \quad \text{s.t.} \quad D_s(x, P) \leq q_s \quad \forall s \in \mathcal{S} \quad (46)$$

MID Continuum Decomposition: The full MID problem couples discrete zoning $x \in \Omega$ with the exact recurrence $R_{g,r} = \min(R_{g,r-1}, e_{g,r})$ across all applicant types and preference ranks. To solve this efficiently, we decompose the problem by maintaining an active prefix vector $\mathbf{k} = (k_g)_{g \in \mathcal{G}}$ with $k_g \in \{0, \dots, R_g\}$.

For each type g , alternatives are partitioned into an exact active prefix ($r < k_g$) and a relaxed tail ($r \geq k_g$). The tail is replaced by an optimistic transportation flow $z_{g,r} \geq 0$:

$$\sum_{r=k_g}^{R_g} z_{g,r} \leq R_{g,k_g-1}, \quad 0 \leq z_{g,r} \leq a_{g,r}(x) \quad \forall g \in \mathcal{G}, r \geq k_g \quad (47)$$

The master problem maximizes the combined welfare over zoning $x \in \Omega$, cutoffs

$P \geq 0$, prefix recurrence variables, and tail flows:

$$\begin{aligned}
& \max_{x, P, R, z} \sum_{g \in \mathcal{G}} n_g \left(\sum_{r < k_g} u(g, r)(R_{g, r-1} - R_{g, r}) + \sum_{r \geq k_g} u(g, r)z_{g, r} \right) \\
& \text{s.t. } x \in \Omega(G, C, k, \delta) \\
& R_{g, r} = \min(R_{g, r-1}, e_{g, r}) \quad \forall g \in \mathcal{G}, r < k_g \\
& \sum_{r \geq k_g} z_{g, r} \leq R_{g, k_g-1}, \quad 0 \leq z_{g, r} \leq a_{g, r}(x) \quad \forall g \in \mathcal{G}, r \geq k_g \\
& \sum_{g, r < k_g: s(g, r)=s} n_g d_{g, r} + \sum_{g, r \geq k_g: s(g, r)=s} n_g z_{g, r} \leq q_s \quad \forall s \in \mathcal{S}
\end{aligned} \tag{48}$$

Proposition 3 (Valid Relaxation). *The master problem (48) is a valid relaxation of the full MID problem, satisfying $\text{OPT}(\text{Full MID}) \leq \text{OPT}(\text{Master}(\mathbf{k}))$.*

Proof. Under any feasible solution to the full MID problem, true tail assignments $d_{g, r} \geq 0$ satisfy $\sum_{r \geq k_g} d_{g, r} = R_{g, k_g-1} - R_{g, R_g} \leq R_{g, k_g-1}$, obey geographic access $d_{g, r} \leq a_{g, r}(x)$, and clear program capacity. Because the master problem relaxes cutoff consistency within tails, the true solution is feasible in the master with identical objective value. ■

Separation and Optimality: Given master candidate $(\bar{x}, \bar{P})$, true demand $D_s(\bar{x}, \bar{P})$ and assigned masses $d_{g, r}(\bar{x}, \bar{P})$ are evaluated using the complete recurrence:

1. **Overload Separation:** If $D_s(\bar{x}, \bar{P}) > q_s$ for any program s , we identify a minimal subset of contributing types whose prefix exact demand violates capacity, and increment k_g through the offending rank, cutting off the candidate.
2. **Utility-Gap Separation:** If capacity is feasible but $\sum_{r \geq k_g} u(g, r)\bar{z}_{g, r} > \sum_{r \geq k_g} u(g, r)d_{g, r}(\bar{x}, \bar{P})$ for some type g , we increment k_g through the first tail rank where $\bar{z}_{g, r} \neq d_{g, r}(\bar{x}, \bar{P})$.

When no separation is possible, complete demand is feasible and true utility equals master utility, proving that candidate $\bar{x}$ is a globally optimal zoning.

6.2 Sample Average Approximation (SAA)

Under strict lottery tiebreakers, student assignment is equivalent to stable matching. Under SAA, we draw independent tie-breaking seeds $\psi \in \Psi$, yielding strict priority orderings $\succ_s^\psi$ and student preferences $\succ_i^\psi$.

For a fixed sample ψ the recourse problem is the fractional stable-admissions program (69)–(74), in which the zoning enters only through the binary access indicators $a_{i, s}$ and stability is imposed by the comb inequalities (73) of Baïou and Balinski (2000). Appendix B develops that polytope in full: the shaft-and-tooth characterization of blocking, the validity and completeness of the comb

family once the set of acceptable pairs is itself a decision variable, the aggregated one-row-per-pair surrogate (76) that the implementation loads in its place, the separation routine that generates combs, and the variant in which zoning confers a priority bonus rather than exclusive access. Three facts established there are used below.

1. Every family in play is valid at every zoning and every seed, so the LP value is an upper bound on sample stable-matching welfare and any cut read off its dual is sound.
2. With the full comb family the recourse value equals the welfare of deferred acceptance exactly, so the outer approximation is tangent at the candidate zoning rather than merely valid there.
3. With the aggregated surrogate it is not. On Block₂ the surrogate over-reports realized deferred-acceptance welfare by 194.8 to 251.8 units across eight saved zonings, mean 225.0, and the variation across zonings is what can misorder two candidates.

The derivation below uses only the first.

Dual Outer Approximation and Geographic Aggregation: By LP duality, the dual solution $(y^{1,\psi,r}, y^{2,\psi,r}, y^{3,\psi,r}, y^{4,\psi,r})$ computed at iteration r under seed ψ gives a valid global upper bound on assignment utility for any choice of student-program access indicators:

$$\eta_\psi \leq \kappa^{(r,\psi)} + \sum_{i \in \mathcal{I}} \sum_{s \in \mathcal{S}} \beta_{i,s}^{(r,\psi)} a_{i,s} \quad (49)$$

where $\kappa^{(r,\psi)} = \sum_{i \in \mathcal{I}} y_i^{1,\psi,r} + \sum_{s \in \mathcal{S}} q_s y_s^{2,\psi,r}$ is the baseline dual welfare intercept, and $\beta_{i,s}^{(r,\psi)} = y_{i,s}^{3,\psi,r} + q_s y_{i,s}^{4,\psi,r}$ collects the multipliers of the two rows in which $a_{i,s}$ appears, so it reads as the marginal dual value of granting student i access to program s ; Appendix B.7 notes that this reading, and the sign that comes with it, are specific to the aggregated relaxation and do not survive the addition of comb rows.

Because school zoning access is determined entirely by geographic residence, an applicant i residing at vertex $v(i) \in V$ has access to program s located at school vertex $l(s) \in V$ if and only if $v(i)$ and $l(s)$ are co-zoned. That is, $a_{i,s} \equiv a_{v(i),l(s)}$ for all non-citywide programs (with $a_{i,s} = 1$ identically for citywide programs or when $v(i) = l(s)$).

Partitioning students by home vertex $\mathcal{I}_v = \{i \in \mathcal{I} \mid v(i) = v\}$ and programs by school vertex $\mathcal{S}_u = \{s \in \mathcal{S} \mid l(s) = u\}$, the student-level dual terms sum to aggregate vertex-pair coefficients:

$$\Gamma_{v,u}^{(r,\psi)} = \sum_{i \in \mathcal{I}_v} \sum_{\substack{s \in \mathcal{S}_u \\ s \text{ non-citywide}}} \beta_{i,s}^{(r,\psi)} \quad \forall v \in V, u \in \mathcal{S}, v \neq u \quad (50)$$

Fixed access terms (citywide programs and students residing at a school vertex) are absorbed directly into the intercept:

$$\tilde{\kappa}^{(r,\psi)} = \kappa^{(r,\psi)} + \sum_{i \in \mathcal{I}} \sum_{\substack{s \in \mathcal{S} \\ s \text{ citywide} \vee v(i)=l(s)}} \beta_{i,s}^{(r,\psi)} \quad (51)$$

Furthermore, rather than introducing separate epigraph variables η_ψ for each sampled lottery seed $\psi \in \Psi$, the master problem optimizes a single expected welfare variable $\eta \in \mathbb{R}$. Averaging across all $|\Psi|$ samples at iteration r yields a single aggregated Benders cut:

$$\eta \leq \bar{\kappa}^{(r)} + \sum_{v \in V} \sum_{u \in S} \bar{\Gamma}_{v,u}^{(r)} a_{v,u} \quad (52)$$

where:

$$\bar{\kappa}^{(r)} = \frac{1}{|\Psi|} \sum_{\psi \in \Psi} \tilde{\kappa}^{(r,\psi)}, \quad \bar{\Gamma}_{v,u}^{(r)} = \frac{1}{|\Psi|} \sum_{\psi \in \Psi} \Gamma_{v,u}^{(r,\psi)} \quad (53)$$

The SAA master problem then optimizes expected welfare over the zoning partition:

$$\max_{x,a,\eta} \quad \eta \quad (54)$$

$$\text{s.t.} \quad x \in \Omega(G, C, k, \delta) \quad (55)$$

$$\eta \leq \bar{\kappa}^{(r)} + \sum_{v \in V} \sum_{u \in S} \bar{\Gamma}_{v,u}^{(r)} a_{v,u} \quad \forall r = 1, \dots, R \quad (56)$$

$$a_{v,u} \iff \bigvee_{z \in [k]} (x_{v,z} \wedge x_{u,z}) \quad \forall (v,u) \in \mathcal{A}^* \quad (57)$$

$$0 \leq \eta \leq \text{Welfare}_{\max} \quad (58)$$

$$a_{v,u} \in \{0, 1\} \quad \forall (v,u) \in \mathcal{A}^* \quad (59)$$

where $\mathcal{A}^* = \{(v,u) \in V \times S \mid \exists r, \bar{\Gamma}_{v,u}^{(r)} \neq 0\}$ denotes the sparse set of active vertex-school pairs with non-zero marginal dual value, and $\text{Welfare}_{\max}$ is a global a priori upper bound developed in Appendix C.

A Priori Welfare Bound: The constant $\text{Welfare}_{\max}$ in (58) is nominally just the domain of η , but it also caps the master's reported dual bound and hence the reported gap. Appendix C.1 develops the two candidates: the first-choice constant $\bar{W}^{\text{FC}}$, which ignores capacity and is therefore loose by the full extent of oversubscription, and the transportation bound $\bar{W}^{\text{TP}}$ of Proposition 4, which respects capacity, remains independent of the zoning, is attained integrally, solves in milliseconds and weakly dominates. We take $\text{Welfare}_{\max} = \bar{W}^{\text{TP}}$; on Block₂ it cuts 598 to 612 units off the constant. What neither constant prices is the zone confinement itself, which is the subject of Section 6.3.

Computational Advantages of Aggregation: This formulation confers three major advantages:

1. **Dimensionality Reduction:** Rather than tracking $|\mathcal{I}| \times |\mathcal{S}| \approx 4.5 \times 10^5$ student-level access indicators across $|\Psi|$ epigraph bounds, the master model instantiates binary access variables $a_{v,u}$ only on-demand for pairs (v, u) with positive dual welfare values $\bar{\Gamma}_{v,u}^{(r)} > 0$.
2. **Exact Equivalence:** Because geographic access $a_{i,s} \equiv a_{v(i),l(s)}$ is strictly identical for all students at vertex v , the algebraic aggregation of dual coefficients preserves exact LP outer-approximation guarantees with zero approximation error.
3. **Unified Base Solver Architecture:** Because the cut (52) shares the identical access variable structure $a_{v,u}$ as the choice set objective (40), SAA directly reuses the core CP-SAT and MIP master solvers without requiring custom master models.

6.3 Congestion-Priced Access

The residual gap left by $\bar{W}^{\text{TP}}$ is the unpriced cost of confining each applicant to the programs co-zoned with their residence. Appendix C.2 closes it by replacing the *constant* $\text{Welfare}_{\max}$ with a *function* of the zoning that is still a valid upper bound on stable-matching welfare. Fix a non-negative price π_s per seat at program s , charge it against utility to obtain the priced value $w_{i,s} = u_{i,s} - \pi_s$, and give each applicant the best priced value they can reach under the zoning, with the outside option available at zero. Summing those per-applicant terms and adding back $\sum_s q_s \pi_s$ gives the bound $\bar{W}^{\text{PA}}(x; \pi)$. It is valid for *every* $\pi \geq 0$ and every seed, so the prices are a free modelling choice rather than a quantity that must be computed correctly; it separates across applicants, so it admits an exact finite closed-form outer description in the master’s access variables with no recourse LP at all; and the resulting cutting plane converges finitely to a certified optimum of the bound. On Block₂ it overstates sample-average stable-matching welfare by 249 to 354 units against the 3,877 to 4,850 of $\bar{W}^{\text{TP}}$, and by nearly the same amount at zonings produced by five unrelated methods — the empirical regularity under which maximizing the bound approximates maximizing welfare. Setting $\pi = 0$ isolates what the pricing contributes: the unpriced surrogate converges faster but is indifferent between a program with spare seats and an oversubscribed one, and gave up 1,357 units of realized welfare on one instance at a matched iteration allowance. Proposition 5, Proposition 6 and Theorem 2, the vertex aggregation, the separation rule and the bias decomposition are all in Appendix C.2.

6.4 Empirical Comparison on Budget Welfare

The three budget-set formulations optimize three different surrogates, so they can only be compared on a measure external to all of them. We use the welfare of the discrete MID program, written W^{MID} and called *budget welfare* below, recomputed for every saved zoning by replaying it through the finite-grid MID

oracle at base-block resolution. This is the only quantity computed identically for every strategy: each strategy’s own reported objective is its own surrogate — the sample average for SAA, the priced bound $\overline{W}^{\text{PA}}$ for priced access, the MNL logsum for choice-set optimization — and comparing those to one another compares different functions. We report neither the logsum nor any strategy’s own objective here for that reason.

Design: All arms solve the same Block₂ instance ($|\mathcal{I}| = 3953$, $|\mathcal{S}| = 123$, $k = 6$) under identical structural constraints, with a 3600s solver budget, an allowance of 30 iterations, and 16 threads, replicated over seeds 1–4 and both centroid configurations: 40 runs, no failures. Each run contributes the single zoning, out of every zoning it saved, with the highest budget welfare — not the one its own surrogate selected. That choice is deliberately generous and is applied uniformly; it removes the confound that the strategies disagree about which of their own iterates to return, and it matters, because 12 of the 39 scored runs held an iterate better than the one they handed back. Priced access is run at three price multipliers, $\pi = 0$, π^* and $2\pi^*$, since Proposition 5 leaves the multiplier free.

Mean W^{MID} and the seed spread, largest minus smallest, over the four seeds:

arm	6-zone-3		6-zone-9	
	mean	spread	mean	spread
Priced access, $2\pi^*$	14,957.9	233.7	14,706.2	269.4
Priced access, π^*	14,932.7	287.5	14,714.9	311.8
Priced access, $\pi = 0$	14,808.4	100.6	14,599.7	303.8
SAA	14,777.7	68.4	14,578.7	92.5
Choice set	14,052.4	1,476.4	14,307.0	38.9

Pairing: The marginal spreads above are not the right yardstick for the differences, because every arm ran the same four seeds and a seed’s difficulty is common to all of them. Differencing within a (centroid, seed) pair removes that shared component. Over the eight pairs, relative to SAA:

arm – SAA	pairs won	mean	range
Priced access, $2\pi^*$	7 / 8	+153.9	−78.7 ... +249.3
Priced access, π^*	6 / 8	+145.6	−32.0 ... +334.9
Priced access, $\pi = 0$	5 / 8	+25.9	−206.5 ... +183.9
Choice set	0 / 7	−465.0	−1,590.3 ... −90.3

Priced access at π^* or $2\pi^*$ improves on SAA by roughly 150 units of budget welfare, about 1%, winning six or seven of eight pairs. We state that margin rather than a larger one because single-seed runs at a shorter budget suggested +300 to +400, and the replication does not support it: the effect is real, consistently signed, and smaller than one seed indicated. It is also not uniform — one or two pairs go the other way.

Three further comparisons are worth separating from that headline. Pairing $2\pi^*$ against π^* gives four wins of eight and a mean of +8.3, so the two multipliers

are indistinguishable here and we make no claim between them. Pairing either against $\pi = 0$ gives +128.0 (seven of eight) and +119.7 (five of eight), so congestion pricing is worth about 130 units — real, but an order of magnitude less than the 1,357 it was worth at a 900s budget, because with enough iterations the unpriced surrogate largely catches up. And SAA is the *most stable* arm by a wide margin, with a seed spread of 68 and 93 against 234 to 312 for priced access: it is beaten on the mean while being the most predictable. Choice-set optimization loses every pair, and its 1,476 spread at 6-zone-3 includes one run that returned no feasible zoning at all within its budget, which is why its count there is three rather than four.

Convergence and cost: The difference in how the two cutting-plane families terminate is larger than the difference in what they find. Across the 24 priced-access runs, the relative optimism gap between the master’s model value and the true value of $\sum_i g_i$ at the same zoning falls from a median of 29.78% at the first iteration to a median of 0.013%, and 23 of the 24 runs stop because *no cut separates* — the exact termination certificate of Theorem 2 — after a median of 10 iterations and 466s, never needing the 30-iteration allowance or the 3600s budget. Across the eight SAA runs the same gap falls from 29.23% only to 7.95%, and every run exhausts its iteration allowance at 30 iterations and 4,555s of wall time. Priced access therefore reaches a certified optimum of its surrogate in roughly a tenth of the wall time SAA spends failing to close its own gap, and the reason is structural rather than incidental: its cuts are closed form, so an iteration costs one master solve, whereas each SAA iteration also pays for $|\Psi| = 15$ recourse LPs.

Two cautions attach to that. First, the certificate is about $\overline{W}^{\text{PA}}$, not about W^{MID} ; the run stops because nothing more can be gained on the surrogate, and the remaining distance to the true optimum is bounded by the surrogate’s bias rather than by anything the run measures. Second, choice-set optimization also closes its gap, to a median of 0.01%, but its surrogate is the MNL logsum, which is neither an upper nor a lower bound on realized welfare, so its gap certifies nothing about the matching; its poor budget welfare despite an almost exactly converged model is the sharpest illustration in this table that converging on the wrong surrogate is not progress.

Compactness does not have to be traded away for this. At 6-zone-3 the priced-access arms return 1,128 to 1,136 cut edges against 1,265 for SAA, so they are simultaneously better on both panels; at 6-zone-9 they range from 1,184 to 1,225 against SAA’s 1,179, which is a wash.

Scope: These are four seeds on one instance family at one district size, so the ordering should be read as replicated on this instance rather than established in general, and the ± 150 margin over SAA is small enough that a different instance could plausibly reverse it. What is not marginal, and what we would expect to transfer, is the termination behaviour: a bound that separates across applicants admits an exact finite cut family and therefore a real stopping certificate, whereas a single dual hyperplane per iteration over $|\mathcal{I}| \times |\mathcal{S}|$ access coordinates does not, independently of which zoning either one happens to return.

A Extended Demographic Diversity Constraints

In addition to socioeconomic (FRL) balance, the optimization framework supports multi-group racial and ethnic demographic balance across an arbitrary set of tracked groups $\mathcal{E}$.

Let $r_{v,e} \geq 0$ denote the count of students of ethnicity $e \in \mathcal{E}$ residing in vertex $v \in V$. The district-wide proportion of group e is:

$$R_e = \frac{\sum_{v \in V} r_{v,e}}{\sum_{v \in V} n_v} \quad (60)$$

For each group $e \in \mathcal{E}$ and each zone $Z_i \in \mathcal{P}$, the localized proportion must not deviate from R_e by more than tolerance $\delta_R \in \mathbb{R}_{>0}$:

$$\left| \frac{\sum_{v \in Z_i} r_{v,e}}{N(Z_i)} - R_e \right| \leq \delta_R \quad \forall e \in \mathcal{E}, \forall Z_i \in \mathcal{P} \quad (61)$$

In the CP-SAT and MIP formulations, this is linearized into upper and lower integer bounds:

$$\sum_{v \in V} [M \cdot (r_{v,e} - (R_e - \delta_R)n_v)] \cdot x_{v,z} \geq 0 \quad \forall e \in \mathcal{E}, \forall z \in [k] \quad (62)$$

$$\sum_{v \in V} [M \cdot (r_{v,e} - (R_e + \delta_R)n_v)] \cdot x_{v,z} \leq 0 \quad \forall e \in \mathcal{E}, \forall z \in [k] \quad (63)$$

which are added seamlessly alongside the base FRL constraints without altering the core solver mechanics.

B The Stable-Matching Access Polytope

This appendix develops the sample recourse problem of Section 6.2. Section B.1 fixes the market and the shaft-tooth apparatus, Section B.2 states the comb family and proves it valid and complete at a fixed access pattern, Section B.3 extends both to a market whose set of acceptable pairs is itself a decision variable — the case the zoning master needs — and Section B.4 gives the separation routine. Section B.5 then replaces exclusive access by a zone *priority bonus*, which is the status quo the district actually operates: residence raises a family’s priority at the programs in its zone without excluding anyone. The same polytope survives once each applicant is split into a zoned and an unzoned copy. Section B.6 gives the corresponding separation and Section B.7 records what changes in the master cuts.

B.1 The Sample Market

Fix a tie-breaking seed $\psi \in \Psi$. Applicants are $\mathcal{I}$, programs are $\mathcal{S}$, and $\Gamma = \{(i, s) : s \text{ appears on } i\text{'s list}\}$ is the set of acceptable pairs; $|\Gamma| = 12,370$ on Block₂ against $|\mathcal{I}| \times |\mathcal{S}| \approx 4.9 \times 10^5$, because ranked lists average 3.13 entries.

Program s has q_s seats and, under seed ψ , a strict order $\succ_s^\psi$ over $\{i : (i, s) \in \Gamma\}$ obtained by sorting the priority tier $\rho_{i,s}$ and breaking ties by the lottery draw $\omega_{i,s}^\psi$. Applicant i has a strict order $\succ_i^\psi$ over their list and utilities $u_{i,s}$ consistent with it, so $s \succ_i^\psi s'$ implies $u_{i,s} \geq u_{i,s'}$. We write $\succeq$ for the reflexive closures. Following the proof of Proposition 4, $\mathcal{S}$ contains an outside option s_0 with $u_{i,s_0} = 0$, $q_{s_0} = |\mathcal{I}|$, universally acceptable and ranked last by everyone, so $\sum_s D_{i,s} = 1$ loses no generality and unassignment is the mass placed on s_0 .

Zoning enters through the access indicators $a_{i,s} \in \{0, 1\}$ of Section 6.2: $a_{i,s} = 1$ when s is citywide, when $l(s) = v(i)$, or when $v(i)$ and $l(s)$ are co-zoned, and $a_{i,s} = 0$ otherwise. It is convenient to name the market that a given access pattern leaves behind,

$$\Gamma(a) = \{(i, s) \in \Gamma : a_{i,s} = 1\}, \quad (64)$$

and to write $n_s(a) = |\{i : (i, s) \in \Gamma(a)\}|$ for the number of applicants who both rank s and may attend it. A pair $(i, s) \in \Gamma(a)$ *blocks* an integral matching D if i prefers s to their assignment and s either has a free seat or has admitted someone below i in $\succ_s^\psi$; D is stable if no pair of $\Gamma(a)$ blocks it.

B.2 Shafts, Teeth and Combs

The description below is that of Baïou and Balinski (2000), in the notation of Sethuraman, Teo and Qian (2006). For $(i, s) \in \Gamma$ define the *shaft* and the *tooth*

$$S^\psi(s, i) = \{(i', s) \in \Gamma : i' \succeq_s^\psi i\}, \quad T^\psi(i, s) = \{(i, s') \in \Gamma : s' \succeq_i^\psi s\}, \quad (65)$$

and for any $\hat{\Gamma} \subseteq \Gamma$ write $D(\hat{\Gamma}) = \sum_{(i,s) \in \hat{\Gamma}} D_{i,s}$. The shaft collects the seats of s that go to applicants s likes at least as much as i ; the tooth collects the outcomes i likes at least as much as s . Both are prefixes of an order and are therefore running sums, which is what makes the whole apparatus cheap: the implementation carries one prefix variable per element of Γ on each side, $2|\Gamma|$ in total, rather than re-summing.

Lemma 1 (Blocking in shaft–tooth form). *Let D be an integral matching and $(i, s) \in \Gamma(a)$. Then (i, s) blocks D if and only if*

$$D(S^\psi(s, i)) \leq q_s - 1 \quad \text{and} \quad D(T^\psi(i, s)) = 0. \quad (66)$$

Proof. The second condition says i receives nothing weakly preferred to s , which is exactly the applicant half of blocking. For the program half, $D(S^\psi(s, i))$ counts the applicants weakly above i admitted to s . If that count is at most $q_s - 1$ then the remaining $q_s - D(S^\psi(s, i)) \geq 1$ seats are either empty or held by applicants strictly below i , so s can accept i , dropping its worst admit if necessary. Conversely if the count is q_s then s is full with applicants it weakly prefers to i and gains nothing from i . Note that i itself is in $S^\psi(s, i)$, so under $D(T^\psi(i, s)) = 0$ the count is unchanged by i 's own row. ■

Lemma 1 is what the comb construction linearizes. A single applicant witnesses only one seat, so certifying that all q_s seats are unavailable to i requires

q_s witnesses. Fix $(i, s) \in \Gamma$ and a collection K of $q_s - 1$ *teeth bases*, distinct except that the vacant slot may repeat, each drawn from

$$\mathcal{K}^\psi(s, i) = \{k \in \mathcal{I} : (k, s) \in \Gamma, k \succ_s^\psi i\} \cup \{\emptyset\}, \quad (67)$$

where $\emptyset$ is a *vacant slot*, explained below. The *comb* based at (i, s) with teeth K is

$$C^\psi(s, i, K) = S^\psi(s, i) \cup (T^\psi(i, s) \setminus \{(i, s)\}) \cup \bigcup_{k \in K} (T^\psi(k, s) \setminus \{(k, s)\}), \quad (68)$$

a shaft with q_s teeth hanging off it, one for the base and one for each element of K , with the convention that a vacant slot contributes no pairs to the union and enters only through its credit. Each tooth is stripped of its own $(\cdot, s)$ entry because that entry already lies in the shaft, so nothing is double counted. The comb inequality asserts that a comb carries at least q_s units of mass. Written out, and with access and vacancies priced as derived in Section B.3,

$$\max_D \sum_{i \in \mathcal{I}} \sum_{s \in \mathcal{S}} u_{i,s} D_{i,s} \quad (69)$$

$$\text{s.t.} \quad \sum_{s \in \mathcal{S}} D_{i,s} = 1 \quad \forall i \in \mathcal{I} \quad (70)$$

$$\sum_{i \in \mathcal{I}} D_{i,s} \leq q_s \quad \forall s \in \mathcal{S} \quad (71)$$

$$D_{i,s} \leq a_{i,s} \quad \forall (i, s) \in \Gamma \quad (72)$$

$$\underbrace{\sum_{i' \succeq_s^\psi i} D_{i',s}}_{\text{shaft}} + \underbrace{\sum_{s' \succ_i^\psi s} D_{i,s'}}_{\text{base tooth}} + \sum_{k \in K} \underbrace{\left[\sum_{s' \succ_k^\psi s} D_{k,s'} + \chi_{k,s} \right]}_{\text{tooth } k} \geq q_s a_{i,s} \quad (73)$$

$$\forall (i, s) \in \Gamma \text{ and all admissible } K$$

$$D_{i,s} \geq 0 \quad \forall (i, s) \in \Gamma \quad (74)$$

where the *tooth credit* is

$$\chi_{k,s} = 1 - a_{k,s} \text{ for } k \in \mathcal{I}, \quad \chi_{\emptyset,s} = 1. \quad (75)$$

The credit is the only departure from the fixed-market description, and Section B.3 is where it is earned. For the moment read (73) at $a \equiv 1$ and with no vacant slots, where every credit is zero and the inequality is literally $D(C^\psi(s, i, K)) \geq q_s$.

Proposition 2 (Comb Inequality Stability). *Fix ψ and a binary access pattern a . Then the integral points of (70)–(74) are exactly the stable matchings*

of the market $\Gamma(a)$, and the system has no fractional extreme points: its solution set is $\text{conv}\{D : D \text{ stable in } \Gamma(a)\}$.

Proof. Validity and integral exactness are Proposition 10 below, which proves them for all a at once. For the absence of fractional extreme points, take $a \equiv 1$ first and assume $n_s \geq q_s$ for every s . Then all credits vanish, (73) is the comb inequality $D(C) \geq q_s$, and the claim is the theorem of Baïou and Balinski (2000): the system (70)–(74) is the stable admissions polytope. Sethuraman, Teo and Qian (2006) give a short proof by exhibiting, for each fractional point, a decomposition of its mass into q_s unit bins per program such that the level sets $\{\text{best } q_s \text{ applicants at height } \alpha\}$ are stable matchings for every $\alpha \in (0, 1]$; the point is then the convex combination of those level sets induced by α uniform. The general case is the second half of Proposition 10. ■

The completeness claim has a consequence worth isolating, because it says what the recourse oracle would return if the comb family were loaded in full.

Corollary 1 (The comb-complete recourse value is deferred-acceptance welfare). *For every zoning x and seed ψ , the optimal value of (69)–(74) equals the utilitarian welfare of the applicant-proposing deferred-acceptance outcome on $\Gamma(a(x))$.*

Proof. By Proposition 2 the feasible set is the convex hull of the stable matchings of $\Gamma(a)$, so the optimum is attained at one of them and equals the largest welfare of a stable matching. Deferred acceptance returns the applicant-optimal stable matching: every applicant weakly prefers it to every other stable matching (Gale and Shapley, 1962; Roth and Sotomayor, 1990). Since $u_{i \cdot}$ is consistent with $\succ_i^\psi$, each applicant’s utility is weakly largest there, hence so is the sum. ■

Corollary 1 is the reason the comb family is worth stating even though the implementation does not load it. An oracle carrying it would be tangent to realized welfare at the candidate zoning, so the Benders cut (52) would be exact at x^r rather than merely valid, and the incumbent selection problem noted in Section 6.2 would not arise.

B.3 Access as a Decision Variable

The zoning master does not solve one market. It searches over x , and each x induces a different set $\Gamma(a(x))$ of acceptable pairs. A description of the stable matchings of a *fixed* market is therefore not enough: the master needs one family of inequalities, written in the variables (D, a) , that is valid simultaneously at every binary a and that specialises to the complete description of $\Gamma(a)$ at each one. Two things obstruct the naive attempt of multiplying the right-hand side by $a_{i,s}$ and leaving the comb otherwise alone.

First, a tooth base without access is not a competitor. Let $q_s = 2$ with applicants $1 \succ_s^\psi 2$, both ranking only s , and let $a_{1,s} = 0$, $a_{2,s} = 1$. The stable matching of $\Gamma(a)$ assigns 2 to s and leaves 1 unassigned, but the uncredited comb based at $(2, s)$ with tooth 1 reads $D_{1,s} + D_{2,s} + 0 + 0 = 1 < 2 = q_s a_{2,s}$. The row is violated by a genuine stable matching, so the family without credits is not valid.

Second, a program may have fewer than q_s interested applicants with access, in which case there are not $q_s - 1$ genuine competitors above i to serve as teeth and, absent the vacant slot, the base (i, s) carries no usable row — yet stability has content there, since a program that cannot fill must admit everyone who wants it. Both obstructions are the same phenomenon and both are repaired by (75): an applicant who cannot compete for s , whether because they lack access or because they do not exist, is credited a full unit, because for the purpose of s 's market they are indistinguishable from an applicant already served better than s .

Proposition 10 (Zoning-robust comb inequalities). *Let ψ be fixed.*

- (i) (Uniform validity.) *Every inequality (73) holds at every pair (D, a) with $a \in \{0, 1\}^\Gamma$ and D an integral stable matching of $\Gamma(a)$.*
- (ii) (Integral exactness.) *Conversely, if a is binary and D is integral and satisfies (70)–(74), then D is a stable matching of $\Gamma(a)$.*
- (iii) (Specialisation.) *At each fixed binary a , the rows of (73) imply Baiou and Balinski's complete description of the market $\Gamma(a)$, padded as required when $n_s(a) < q_s$. Hence the solution set is exactly $\text{conv}\{D : D \text{ stable in } \Gamma(a)\}$, completing Proposition 2.*

Proof. (i) If $a_{i,s} = 0$ the right-hand side is zero and the left-hand side is a sum of non-negative terms. So assume $a_{i,s} = 1$, fix teeth K , and write $P = \{i\} \cup K$, a collection of q_s elements, and

$$b_i = \sum_{s' \succ_i^\psi s} D_{i,s'}, \quad b_k = \sum_{s' \succ_k^\psi s} D_{k,s'} + \chi_{k,s} \quad (k \in K),$$

so that the left-hand side of (73) is $D(S^\psi(s, i)) + \sum_{p \in P} b_p$. Suppose it is at most $q_s - 1$. Partition P into

$$G = \{p \in P : b_p \geq 1\}, \quad F = \{p \in P \setminus G : D_{p,s} = 1\}, \quad H = P \setminus (F \cup G).$$

Every $p \in F$ has $b_p = 0$, hence $\chi_{p,s} = 0$, hence $a_{p,s} = 1$ and $p \succeq_s^\psi i$ — vacant slots and applicants without access carry a credit and lie in G — so $(p, s) \in S^\psi(s, i)$ and $D(S^\psi(s, i)) \geq |F|$. Therefore the left-hand side is at least $|F| + |G|$, and since $|F| + |G| + |H| = q_s$ the assumed bound forces $H \neq \emptyset$. Pick $k \in H$. As above $\chi_{k,s} = 0$, so $(k, s) \in \Gamma(a)$ and $k \succeq_s^\psi i$; and $b_k = 0$ with $D_{k,s} = 0$ gives $D(T^\psi(k, s)) = 0$. Finally $k \succeq_s^\psi i$ gives $S^\psi(s, k) \subseteq S^\psi(s, i)$, so

$$D(S^\psi(s, k)) \leq D(S^\psi(s, i)) \leq q_s - 1.$$

By Lemma 1 the pair $(k, s) \in \Gamma(a)$ blocks D , contradicting stability.

(ii) Let a be binary and D integral and feasible. Constraints (70), (71) and (72) make D a matching of $\Gamma(a)$. Suppose $(i, s) \in \Gamma(a)$ blocked it, so by Lemma 1 $D(T^\psi(i, s)) = 0$ and $D(S^\psi(s, i)) \leq q_s - 1$. Let $A = \{k : (k, s) \in \Gamma, k \succ_s^\psi i\}$ and, with b_k as in part (i), let $m_0 = |\{k \in A : b_k = 0\}|$. Choose the teeth greedily,

taking the $q_s - 1$ smallest brackets of $\mathcal{K}^\psi(s, i)$; since vacant slots have bracket exactly 1 and are available without limit,

$$\sigma := \sum_{k \in K} b_k \leq \max\{0, q_s - 1 - m_0\}.$$

The base tooth vanishes because $D(T^\psi(i, s)) = 0$, so the left-hand side of (73) is $D(S^\psi(s, i)) + \sigma$. Now $D_{i,s} = 0$, so $D(S^\psi(s, i))$ counts the members of A admitted to s ; each of those has access and nothing strictly preferred to s , hence bracket 0, hence $D(S^\psi(s, i)) \leq m_0$. If $m_0 \geq q_s - 1$ then $\sigma = 0$ and the left-hand side is $D(S^\psi(s, i)) \leq q_s - 1$; otherwise it is at most $m_0 + (q_s - 1 - m_0) = q_s - 1$. Either way it falls below $q_s = q_s a_{i,s}$ and this row of (73) is violated, contradicting feasibility. So no pair of $\Gamma(a)$ blocks D .

(iii) Fix binary a . A program with $n_s(a) \geq q_s$ needs no padding, so consider one with $n_s(a) < q_s$ and pad $\Gamma(a)$ with $\varphi_s = q_s - n_s(a)$ phantom applicants whose only acceptable program is s and who occupy the top φ_s positions of $\succ_s^\psi$, each with utility zero. In every stable matching of the padded market each phantom is admitted to s : were one rejected, s would have to be full, but the admitted set draws from $n_s(a)$ real applicants and $\varphi_s - 1$ other phantoms, at most $q_s - 1$ in all. Real applicants therefore face effective capacity $n_s(a)$, which does not bind, so the stable matchings of the padded market restrict exactly onto those of $\Gamma(a)$ and carry the same welfare. Baïou and Balinski's description applies to the padded market, in which $n_s \geq q_s$ for every program. Eliminate the phantom variables, all of which are fixed at one: a padded comb based at (i, s) using j phantom teeth has left-hand side equal to the real shaft plus φ_s — every phantom lies in the shaft, whether or not it was chosen as a tooth — plus the base tooth plus the $q_s - 1 - j$ real tooth terms, since a phantom's own tooth is empty. The row of (73) with the same real teeth and j vacant slots has the same right-hand side and a left-hand side smaller by $\varphi_s - j \geq 0$, hence implies it. Applicants without access are handled identically, being vacant slots of $\Gamma(a)$ under another name: their credit is 1 and their shaft contribution is 0 by (72). So every inequality of the complete description is implied, giving solution set $\subseteq \text{conv}$; part (i) gives $\supseteq$. ■

What the implementation loads. The comb family has up to $\binom{n_s-1}{q_s-1}$ rows per base pair and is separated, not enumerated. What `SaaOracle` loads up front is one aggregated stability row per acceptable pair,

$$\sum_{i' \succ_s^\psi i} D_{i',s} + q_s \sum_{s' \succeq_i^\psi s} D_{i,s'} \geq q_s a_{i,s} \quad \forall (i, s) \in \Gamma, \quad (76)$$

the many-to-one aggregation of Rothblum's (1992) inequality. It is valid: if $a_{i,s} = 1$ and i receives nothing weakly preferred to s then the second sum is zero and stability forces q_s seats to applicants strictly above i , while if i does receive something weakly preferred the second sum is 1 and alone supplies q_s . Being valid, it is implied by the complete description of Proposition 2 and may be added freely. It is also cheap: with the prefix variables of (65) it is $O(|\Gamma|)$ rows, 12,370 on Block₂, and the y^4 block of the dual in Section 6.2 is precisely

its multiplier vector. Use the *strict* shaft as written; the weak variant is valid too but measurably looser, because it lets i 's own mass at s be counted twice.

What (76) is not is complete, and the reason is visible in the derivation. Summing the per-seat inequalities of a seat-cloned program and bounding $\sum_j (q_s - j + 1)D_{i,s(j)}$ by $q_s D_{i,s}$, where $s^{(1)}, \dots, s^{(q_s)}$ are the clones, discards the distinction between one applicant holding a whole seat and several holding fractions of it; the multiplier q_s on the own-preference term is what lets a fractional $D_{i,s}$ pay for a deficit in the shaft. Concretely, let $q_s = 2$ and let applicants $1 \succ_s^\psi 2 \succ_s^\psi 3$ rank only s , with $u_{1,s} = u_{2,s} = 1$ and $u_{3,s} = 10$, all access granted. The unique stable matching admits 1 and 2, for welfare 2. The point $D_{1,s} = 1$, $D_{2,s} = D_{3,s} = \frac{1}{2}$ satisfies (70)–(72), (74) and every row of (76) — the three rows read $0 + 2 = 2$, $1 + 1 = 2$ and $\frac{3}{2} + 1 = \frac{5}{2}$ against $q_s = 2$ — and has objective 6.5, more than three times the truth. The comb based at $(2, s)$ with the single tooth 1 is what removes it: its left-hand side is $D_{1,s} + D_{2,s} = \frac{3}{2} < 2$.

The gap is not merely theoretical. Because Corollary 1 identifies the comb-complete value with deferred-acceptance welfare, the discrepancy between the surrogate LP value and a deferred-acceptance run measures exactly the slack of (76). Over eight saved zonings on Block₂ with $k = 6$ the surrogate over-reports realized deferred-acceptance welfare by 194.8 to 251.8 units, mean 225.0, against a total welfare near 14,500; measured instead against the MID continuum value on the same zonings the over-report is a very stable 194.8 to 195.0. The variation of roughly 57 units *across* zonings is the operationally relevant part, since it is what can reorder two candidates: the cuts stay valid, but a strategy that selects its incumbent by surrogate value can select wrongly by that much. This is why Section 6.2 reports budget welfare from an independent replay rather than from the oracle.

B.4 Comb Separation Under Access

Separation is exact and requires no optimization beyond a sort. Fix a candidate zoning x^r , hence a binary a , and a fractional $\bar{D}$ solving the current restricted LP. For $(k, s) \in \Gamma$ let

$$\bar{b}_{k,s} = \sum_{s' \succ_k^\psi s} \bar{D}_{k,s'} + (1 - a_{k,s}), \quad (77)$$

the bracket of (73); all $|\Gamma|$ of these are available as the tooth prefix sums already in the model. Because the teeth enter (73) additively and their admissible set $\mathcal{K}^\psi(s, i)$ depends on the base only through $k \succ_s^\psi i$, the most violated comb based at (i, s) is obtained by choosing the $q_s - 1$ smallest brackets above i :

$$\text{viol}(i, s) = q_s a_{i,s} - \left[\underbrace{\sum_{i' \succ_s^\psi i} \bar{D}_{i',s}}_{\text{shaft prefix}} + \sum_{s' \succ_i^\psi s} \bar{D}_{i,s'} + \underbrace{\min_{\substack{K \subseteq \mathcal{K}^\psi(s,i) \\ |K|=q_s-1}} \sum_{k \in K} \bar{b}_{k,s}}_{\sigma(i,s)} \right], \quad (78)$$

and vacant slots, whose bracket is 1, enter this minimum on the same footing as everything else — they are selected exactly when fewer than $q_s - 1$ applicants

above i have a bracket below 1, which is the case Section B.3 introduced them for. One descent of $\succ_s^\psi$ computes (78) at every base simultaneously. Carry two accumulators: the shaft, a running sum, and σ , the sum of a max-heap of size $q_s - 1$ holding the smallest brackets seen so far, initialized with $q_s - 1$ vacant slots. On reaching applicant i , first add $\bar{D}_{i,s}$ to the shaft — the shaft of (65) is weak and includes i 's own row — then evaluate (78), then push $\bar{b}_{i,s}$ and pop the maximum, so that on reaching the next applicant the heap holds exactly the teeth strictly above them. Popping the maximum evicts a vacant slot only when a real bracket is smaller, which is why brackets above 1 — possible for an applicant who lacks access and is served better than s — never displace a slot. The cost is $O(|\Gamma| + \sum_s n_s \log q_s)$, dominated by $O(|\Gamma| \log \max_s q_s)$, which on Block₂ is a few hundred thousand operations against an LP re-solve; separation is free relative to the oracle it feeds.

The oracle is then a cutting plane in its own right, nested inside the zoning cutting plane: load (70)–(72), (74) and (76), solve, run the descent above, add every comb with $\text{viol}(i, s) > \epsilon$, and repeat until none is found. Termination is finite because the comb family is finite and no row is added twice, and on exit Proposition 2 gives a value equal to deferred-acceptance welfare by Corollary 1. Two implementation notes. Rows should be added in batches — all violated bases at once — since the marginal cost of a row is small next to a re-solve. And the emitted row must record its teeth as *indices*, not as the numeric bracket values that selected them: the row is an inequality in (D, a) valid at every zoning, and re-deriving it from $\bar{D}$ at the next master iterate would silently produce a different, weaker row.

B.5 Zoning as a Priority Bonus

Exclusive access is the sharpest form a zone can take, but it is not the form the district operates. In the status quo a family receives a priority *boost* at its attendance-area program and may still apply anywhere; the zone changes who wins a contested seat, not who may enter the contest. This subsection writes the polytope for that policy. The construction is the natural one: every applicant is split into a *zoned* and an *unzoned* copy, the priority structure is written once for each copy, and the zoning decides which copy is real.

Data. Let $\rho_{i,s} \in \mathbb{Z}_{\geq 0}$ be the base priority score of i at s , namely the sum of the configured tier weights — CTIP, sibling, pre-K, round and the rest — with the zone term removed; let $\beta \in \mathbb{Z}_{\geq 1}$ be the zone weight, the single policy dial of this subsection; and let $\omega_{i,s}^\psi \in (0, 1)$ be the lottery draw under seed ψ , with $\omega_{i,s}^\psi \neq \omega_{j,s}^\psi$ whenever $i \neq j$. Single and multiple tie-breaking differ only in whether $\omega_{i,s}^\psi$ depends on s , and nothing below uses either. Program s orders applicants by decreasing realized score $\rho_{i,s} + \beta a_{i,s} + \omega_{i,s}^\psi$. Integrality of ρ and β is used only to keep realized scores separated and may be replaced by the requirement that $\max_{i,s} \omega_{i,s}^\psi$ be smaller than the least positive gap between distinct values of $\rho_{i,s} + \beta a_{i,s}$.

The bonus indicator. $a_{i,s} = 1$ when $l(s)$ and $v(i)$ are co-zoned, which includes

$l(s) = v(i)$. This is the same quantity as in the access model and the same master variable $a_{v(i),l(s)}$ of (57), so the zoning side of the problem is untouched: only the recourse changes. Two families of pairs are constant rather than variable and must be fixed as data. The outside option carries no bonus, $a_{i,s_0} \equiv 0$. For a citywide program we take residence in the zone containing it to earn no bonus either, $a_{i,s} \equiv 0$. The access model instead fixes $a_{i,s} = 1$ there, but only to make such programs universally reachable; under a bonus the requirement is merely that the value be constant, since a uniform shift of one program's scores leaves its order unchanged, so either constant serves. Letting $a_{i,s}$ be the co-zoning indicator at citywide programs as well is a different *policy*, not a different convention, and is covered by everything below without modification.

Each applicant carries two *keys* at each program,

$$\kappa_{i,s}^{\text{in}} = \rho_{i,s} + \omega_{i,s}^{\psi} + \beta, \quad \kappa_{i,s}^{\text{out}} = \rho_{i,s} + \omega_{i,s}^{\psi}, \quad (79)$$

of which the zoning selects one: the realized score of i at s is $\kappa_{i,s}^{\text{in}}$ if $a_{i,s} = 1$ and $\kappa_{i,s}^{\text{out}}$ otherwise. Both keys are data; the selection is the decision.

Merged key order. Fix s and let $n_s = |\{i : (i, s) \in \Gamma\}|$. The $2n_s$ keys (79) are pairwise distinct, since equality of any two forces two lottery draws to differ by an integer and hence to coincide. Sort them once into strictly decreasing order and write $\text{pos}_s(i, z) \in \{1, \dots, 2n_s\}$ for the position of $\kappa_{i,s}^z$ and $(i_s(m), z_s(m))$ for the applicant and tier occupying position m . Every zoning realizes exactly one key per applicant, so the realized order $\succ_s^{\psi,a}$ is the subsequence of the merged order at the realized positions and, writing $z(\cdot)$ for realized tiers,

$$k \succ_s^{\psi,a} i \iff \text{pos}_s(k, z(k)) < \text{pos}_s(i, z(i)). \quad (80)$$

Programs whose $a_{\cdot,s}$ is constant — citywide and the outside option — carry n_s live keys rather than $2n_s$. Because no pair is ever withdrawn under a bonus, n_s is data: the vacant slots of Section B.3 are needed at exactly the programs with $n_s < q_s$, and there are $\max\{0, q_s - n_s\}$ of them, independent of the zoning.

Split the assignment variables to match:

$$D_{i,s} = D_{i,s}^{\text{in}} + D_{i,s}^{\text{out}}, \quad 0 \leq D_{i,s}^{\text{in}} \leq a_{i,s}, \quad 0 \leq D_{i,s}^{\text{out}} \leq 1 - a_{i,s}. \quad (81)$$

Lemma 2 (Tier split exactness). *At every binary a and every point satisfying (81), $D_{i,s}^{\text{in}} = a_{i,s} D_{i,s}$ and $D_{i,s}^{\text{out}} = (1 - a_{i,s}) D_{i,s}$, with no integrality required of D .*

Proof. If $a_{i,s} = 1$ the second bound forces $D_{i,s}^{\text{out}} = 0$; if $a_{i,s} = 0$ the first forces $D_{i,s}^{\text{in}} = 0$. ■

Lemma 2 is the whole point of the split. Under a bonus the shaft of a comb is a sum over an index set that *depends on the decision* — who is above i at s is a function of a — and such a sum is bilinear in (a, D) . The split linearizes it exactly, because the products $a_{i,s} D_{i,s}$ that the bilinear form needs are now variables in their own right. Define, for a base pair $(i, s) \in \Gamma$ and a tier $z \in \{\text{in}, \text{out}\}$, the *tier shaft*

$$\Sigma_{s,i}^z = \sum_{(i',s) \in \Gamma} \left(\mathbb{1}[\kappa_{i',s}^{\text{in}} \geq \kappa_{i,s}^z] D_{i',s}^{\text{in}} + \mathbb{1}[\kappa_{i',s}^{\text{out}} \geq \kappa_{i,s}^z] D_{i',s}^{\text{out}} \right), \quad (82)$$

a fixed linear form whose indicators are data: each is evaluated once from (79). Write $\succ_s^{\psi,a}$ for the order the realized scores induce and $S^{\psi,a}(s,i)$ for the shaft (65) taken under it. By Lemma 2, if i 's realized tier is z then $\Sigma_{s,i}^z = D(S^{\psi,a}(s,i))$, and the two forms $\Sigma_{s,i}^{\text{in}}, \Sigma_{s,i}^{\text{out}}$ together cover both possibilities. The identity is exact and not a relaxation: $\Sigma_{s,i}^z$ sums *both* copies of every applicant with a key above position $\text{pos}_s(i,z)$, and Lemma 2 zeroes the copy the zoning did not realize. In particular $\Sigma_{s,i}^{\text{out}}$ contains $D_{i,s}^{\text{in}}$, because $\kappa_{i,s}^{\text{in}} > \kappa_{i,s}^{\text{out}}$; that term vanishes at every zoning where the row's right-hand side is active.

Do not instantiate (82) as written. Each form has $O(n_s)$ nonzeros and there are $2n_s$ of them per program, which costs $O(\sum_s n_s^2)$ nonzeros. Because the forms are nested prefixes of one order, introduce instead $2n_s + 1$ prefix variables per program,

$$E_{s,0} = 0, \quad E_{s,m} = E_{s,m-1} + D_{i_s(m),s}^{z_s(m)} \quad m = 1, \dots, 2n_s, \quad (83)$$

so that $E_{s,m}$ is the mass at s held by the first m positions of the merged order, and read the shafts off it:

$$\Sigma_{s,i}^z = E_{s, \text{pos}_s(i,z)}, \quad \text{and the strict shaft is} \quad E_{s, \text{pos}_s(i,z)-1}. \quad (84)$$

This is $2|\Gamma|$ extra columns and $2|\Gamma|$ extra rows in total, and it leaves every comb row with $O(q_s)$ nonzeros. Do the same on the applicant side, where the bonus changes nothing: let

$$t_{i,s} = \sum_{s' \succ_i^{\psi} s} D_{i,s'}, \quad D_{i,s'} = D_{i,s'}^{\text{in}} + D_{i,s'}^{\text{out}}, \quad (85)$$

be the $|\Gamma|$ tooth prefix variables of (65). Writing $s^{(1)} \succ_i^{\psi} \dots \succ_i^{\psi} s^{(R_i)}$ for i 's list, they are defined by the recursion $t_{i,s^{(1)}} = 0$ and $t_{i,s^{(r)}} = t_{i,s^{(r-1)}} + D_{i,s^{(r-1)}}$, so that $T^{\psi}(i,s)$ carries mass $t_{i,s} + D_{i,s}$.

That leaves the eligibility of the teeth, which is where the two keys reappear. A candidate tooth base k can stand above i at some zoning only if its zoned key does, so the admissible set is

$$\mathcal{K}^z(s,i) = \{k \neq i : (k,s) \in \Gamma, \text{pos}_s(k, \text{in}) < \text{pos}_s(i,z)\} \cup \{\emptyset\}, \quad (86)$$

and the credit records whether k stands above i unconditionally or only by virtue of its own bonus — one integer comparison per tooth:

$$\chi_{i,k,s}^z = \begin{cases} 0 & \text{if } \text{pos}_s(k, \text{out}) < \text{pos}_s(i,z) \quad (k \text{ ahead in either tier}), \\ 1 - a_{k,s} & \text{if } \text{pos}_s(k, \text{out}) > \text{pos}_s(i,z) \quad (k \text{ ahead only when zoned}), \\ 1 & \text{if } k = \emptyset. \end{cases} \quad (87)$$

Collecting everything, the priority-bonus recourse program for sample ψ is

$$\max \sum_{(i,s) \in \Gamma} u_{i,s} (D_{i,s}^{\text{in}} + D_{i,s}^{\text{out}}) \quad (88)$$

$$\text{s.t.} \quad \sum_{s: (i,s) \in \Gamma} (D_{i,s}^{\text{in}} + D_{i,s}^{\text{out}}) = 1 \quad \forall i \in \mathcal{I} \quad (89)$$

$$\sum_{i: (i,s) \in \Gamma} (D_{i,s}^{\text{in}} + D_{i,s}^{\text{out}}) \leq q_s \quad \forall s \in \mathcal{S} \quad (90)$$

$$D_{i,s}^{\text{in}} \leq a_{i,s}, \quad D_{i,s}^{\text{out}} \leq 1 - a_{i,s} \quad \forall (i,s) \in \Gamma \quad (91)$$

$$E_{s,0} = 0, \quad E_{s,m} = E_{s,m-1} + D_{i_s(m),s}^{z_s(m)} \quad \forall s, m \leq 2n_s \quad (92)$$

$$t_{i,s(1)} = 0, \quad t_{i,s(r)} = t_{i,s(r-1)} + D_{i,s(r-1)} \quad \forall i, 2 \leq r \leq R_i \quad (93)$$

$$E_{s, \text{pos}_s(i,z)} + t_{i,s} + \sum_{k \in K} [t_{k,s} + \chi_{i,k,s}^z] \geq q_s \alpha_{i,s}^z \quad \forall (i,s), z, K \quad (94)$$

$$D_{i,s}^{\text{in}}, D_{i,s}^{\text{out}} \geq 0 \quad \forall (i,s) \in \Gamma \quad (95)$$

where (94) ranges over $(i,s) \in \Gamma$, both tiers z , and every collection K of $q_s - 1$ elements of $\mathcal{K}^z(s,i)$ admissible in the sense of Section B.2, and where

$$\alpha_{i,s}^{\text{in}} = a_{i,s}, \quad \alpha_{i,s}^{\text{out}} = 1 - a_{i,s} \quad (96)$$

switch each row on precisely at the zonings where its tier is the realized one. Four points of practice. The access constraint (72) is gone: under a bonus no pair is ever withdrawn, and the tier bounds (91) carry all of the zoning's influence. Those bounds are written as rows rather than as variable bounds for the same reason (72) is in `SaaOracle`: their multipliers are what the master cut of Section 6.2 reads. Every left-hand side is written in the prefix variables (83) and (85), so a comb row has $q_s + 1$ nonzeros plus the credits, whatever n_s is. And Corollary 1 transfers verbatim, since at a fixed zoning (88)–(95) is an ordinary stable admissions market: solved comb-complete, its value is the welfare of deferred acceptance run under the bonus-induced priorities.

As a warm start, the aggregated surrogate of (76) transfers by substituting the strict shaft of (84),

$$E_{s, \text{pos}_s(i,z)-1} + q_s(t_{i,s} + D_{i,s}) \geq q_s \alpha_{i,s}^z \quad \forall (i,s) \in \Gamma, z \in \{\text{in}, \text{out}\}, \quad (97)$$

which is $2|\Gamma|$ rows and is valid because at the active tier it is exactly (76) for the realized market, the unrealized copies inside the prefix being zero by Lemma 2. Counting columns, $2|\Gamma|$ assignment variables, $2|\Gamma|$ shaft prefixes and $|\Gamma|$ tooth prefixes give $5|\Gamma| = 61,850$ on Block₂; counting rows, $|\mathcal{I}| + |\mathcal{S}| + 5|\Gamma| \approx 6.6 \times 10^4$ before a single comb is separated. Lemma 2 fixes one of each variable pair to zero at any binary a , so an oracle solved at a fixed candidate loses half the assignment columns in presolve.

Proposition 11 (Priority-bonus polytope). *Fix ψ and $\beta \geq 1$. The system (88)–(95) is valid at every binary a simultaneously, its integral points are exactly*

the stable matchings under the bonus-induced order, and at each fixed binary a its solution set is their convex hull.

Proof. Fix a binary a and let $\succ_s^{\psi,a}$ be the induced strict order. Exactly one tier $z(i)$ has $\alpha_{i,s}^z = 1$; rows of the other tier have right-hand side zero and non-negative left-hand side. For the active tier, Lemma 2 gives $\Sigma_{s,i}^{z(i)} = D(S^{\psi,a}(s, i))$, and (87) gives $\chi_{i,k,s}^{z(i)} = 0$ exactly when $k \succ_s^{\psi,a} i$ and $\chi_{i,k,s}^{z(i)} = 1$ otherwise, since a tooth with $\kappa_{k,s}^{\text{out}} < \kappa_{i,s}^z \leq \kappa_{k,s}^{\text{in}}$ outranks i if and only if $a_{k,s} = 1$. So the active rows are precisely the inequalities (73) for the ordinary stable admissions market $(\Gamma, \succ_s^{\psi,a})$ in which every pair is acceptable, and Proposition 10 applies to that market verbatim, with credited and vacant teeth playing the role they played there. ■

Three limits of β are worth reading off, because they place the two formulations of this appendix on one axis.

1. $\beta = 0$. The two keys coincide, so $\Sigma^{\text{in}} = \Sigma^{\text{out}}$ and every credit vanishes, the second branch of (87) becoming vacuous. The two tiered rows at a base then have identical left-hand sides and right-hand sides $q_s a_{i,s}$ and $q_s(1 - a_{i,s})$, so at every binary a they jointly assert only that the left-hand side is at least q_s — the zoning-free inequality. The zoning has left the formulation entirely, as it should: with no bonus and no access restriction, welfare does not depend on x .
2. $\beta > \max_{i,s} \rho_{i,s}$, a dominant bonus. Every zoned applicant outranks every unzoned one, so $\succ_s^{\psi,a}$ is the base order restricted to the zone, concatenated with the base order restricted to its complement, and (82) collapses to $\Sigma_{s,i}^{\text{in}} = \sum_{i' \preceq_s^0 i} D_{i',s}^{\text{in}}$ and $\Sigma_{s,i}^{\text{out}} = \sum_{i'} D_{i',s}^{\text{in}} + \sum_{i' \succ_s^0 i} D_{i',s}^{\text{out}}$, where $\preceq_s^0$ is the base order: an unzoned applicant is behind the whole zoned block.
3. Hard access. Add $D_{i,s}^{\text{out}} = 0$ and $\alpha_{i,s}^{\text{out}} = 0$ for every non-citywide s , which declares the unzoned copy *unacceptable* rather than merely low-priority. Then (81) reduces to $D_{i,s} \leq a_{i,s}$, which is (72); with a dominant bonus every tooth satisfies $\kappa_{k,s}^{\text{out}} < \kappa_{i,s}^{\text{in}}$, so (87) reduces to $\chi_{k,s} = 1 - a_{k,s}$, which is (75); and (94) becomes (73). The access polytope of Section B.2 is the face of the priority-bonus polytope on which the unzoned copies vanish.

B.6 Comb Separation Under a Priority Bonus

At a candidate zoning the separation problem is not merely similar to the one under access, it is the same problem. Once a is fixed, only the tier with $\alpha_{i,s}^z = 1$ need be separated, since the other has right-hand side zero, and every credit collapses: for a base (i, s) and a tooth k realized above i , either k is realized unzoned, in which case $\text{pos}_s(k, \text{out}) < \text{pos}_s(i, z)$ and (87) gives 0, or k is realized zoned, in which case both branches give 0 because $a_{k,s} = 1$. Conversely an admissible tooth that is *not* realized above i has $a_{k,s} = 0$ and credit 1, so its bracket is at least 1 and a vacant slot is never worse. The most violated

comb may therefore be sought over the realized competitors alone, with base-independent brackets

$$\bar{b}_{k,s} = \bar{t}_{k,s} \quad (\text{the tooth prefix (85) evaluated at } \bar{D}), \quad (98)$$

and the descent of Section B.4 applies verbatim: walk the merged order at s , skipping the n_s unrealized positions; carry $E_{s,\cdot}$ as the running shaft; carry a max-heap of size $q_s - 1$ over (98), seeded with $q_s - 1$ vacant slots. The cost is again $O(|\Gamma| \log \max_s q_s)$.

What changes is the *lifting*. The descent sees one order, but the row it emits has to hold at every zoning, and the two are reconciled by recording the row's combinatorics rather than the order that produced it. Three items must be lifted, and each corresponds to a way the row would otherwise be tight at x^r and invalid elsewhere.

- **The base tier.** Record which of $a_{i,s}$ or $1 - a_{i,s}$ is the right-hand side multiplier. A row emitted with the constant q_s would remain in force at zonings that move i to the other tier, where its shaft is the wrong prefix.
- **The shaft.** Record it as the single prefix variable $E_{s, \text{pos}_s(i,z)}$ of (84). Expanding the realized prefix into coefficients on the aggregate $D_{i',s}$ would freeze the current order into the row; the prefix formulation removes the temptation, since the shaft is one column reference and the merged order lives in (92) where it belongs.
- **The tooth credits.** For each selected tooth, compare $\text{pos}_s(k, \text{out})$ against $\text{pos}_s(i, z)$ and store the constant 0 or the affine term $1 - a_{k,s}$ accordingly. This is the one place where the collapse noted above is a trap rather than a convenience: at the candidate the two are numerically indistinguishable, both evaluating to 0, so a row that stores the evaluated credit looks correct and is not. Recorded as 0, a tooth that outranks i only when zoned yields a row violated by legitimate stable matchings at any zoning that unzones k — the failure mode of the two-applicant example in Section B.3.

The practical consequence is that a base pair contributes two row families and that at any one candidate only one of them is separated, so the per-iteration row count is what it was under hard access; both families accumulate over iterations as the master moves applicants between tiers.

B.7 Consequences for the Master Cuts

Two consequences of this appendix bear on the formulations in the body.

First, the sign of the cut coefficients. The coefficient of $a_{i,s}$ in the dual cut of Section 6.2 collects the multiplier of (72), which is non-negative, together with q_s times the multipliers of the stability rows based at (i, s) — and, once combs are separated, -1 times the multiplier of every comb row in which i appears as a *credited tooth*, since $a_{i,s}$ enters those rows through $\chi_{i,s} = 1 - a_{i,s}$ on the left. With only the aggregated rows (76) loaded, $a_{i,s}$ occurs in exactly one row and

on one side, and the aggregate $\bar{\Gamma}_{v,u}^{(r)}$ inherits a single sign. With combs, or with the tiered rows (94), it does not. The sparse instantiation of the master must therefore key on $\bar{\Gamma}_{v,u}^{(r)} \neq 0$, as $\mathcal{A}^*$ in (57) does, and the reading of $\beta_{i,s}^{(r,\psi)}$ as a marginal *value* of access is specific to the aggregated relaxation.

Second, and more consequentially, a priority bonus does not admit the congestion-priced bound developed in Appendix C.2. That construction starts from $T(x)$, the transportation value obtained by keeping the access constraints (72) and deleting stability. Under a bonus there is no access constraint to keep. Every zoning has $\Gamma(a) = \Gamma$, so $T(x)$ collapses to the constant $\bar{W}^{\text{TP}}$, the bound $\bar{W}^{\text{PA}}(x; \pi)$ stops depending on x , and its maximizer is arbitrary. The reason is structural rather than technical. Under hard access the zoning acts on welfare through the *feasible set*, which a transportation relaxation still sees; under a priority bonus it acts on welfare exclusively through *who wins a contested seat*, which lives entirely in the stability constraints. Any relaxation that discards stability discards the whole dependence on the decision. The same argument disposes of the monotonicity that Proposition 6 relies on: granting i the bonus at s weakly helps i and weakly harms every applicant that i thereby overtakes, so welfare is not monotone in a and the threshold family (113) has no counterpart. Of the three budget-set formulations, the comb-based recourse is the only one whose dependence on the zoning is of the right kind to price a priority bonus.

C Welfare Bounds for the Budget-Set Masters

This appendix develops the two upper bounds that the SAA master of Section 6.2 can put in the epigraph domain (58): a zoning-independent constant, and a function of the zoning that prices the cost of confinement. The second is the congestion-priced bound the `priced_access` strategy optimizes.

C.1 The A Priori Transportation Bound

The constant $\text{Welfare}_{\max}$ in (58) is merely the domain of the epigraph variable η , but it carries more weight than that role suggests. The optimality gap reported by the master is measured against its dual bound, and that dual bound is the smaller of $\text{Welfare}_{\max}$ and whatever the accumulated cuts (56) certify. Until the cuts descend below the constant, the constant *is* the reported gap.

The natural choice, obtained by granting every applicant their top-ranked program,

$$\bar{W}^{\text{FC}} = \sum_{i \in \mathcal{I}} u_{i,1}, \quad (99)$$

ignores capacity entirely and is therefore loose by the full extent of oversubscription. We replace it with a bound that respects capacity while remaining independent of the zoning.

Proposition 4 (Transportation Bound). *Let*

$$\overline{W}^{\text{TP}} = \max_y \sum_{i \in \mathcal{I}} \sum_{s \in \mathcal{S}} u_{i,s} y_{i,s} \quad (100)$$

$$\text{s.t.} \quad \sum_{s \in \mathcal{S}} y_{i,s} \leq 1 \quad \forall i \in \mathcal{I} \quad (101)$$

$$\sum_{i \in \mathcal{I}} y_{i,s} \leq q_s \quad \forall s \in \mathcal{S} \quad (102)$$

$$y_{i,s} \geq 0 \quad \forall i, s. \quad (103)$$

Then for every zoning $x \in \Omega(G, C, k, \delta)$ and every tie-breaking seed $\psi \in \Psi$, the stable-matching welfare $W_\psi(x)$ satisfies

$$W_\psi(x) \leq \overline{W}^{\text{TP}} \leq \overline{W}^{\text{FC}}. \quad (104)$$

Proof. Let D be the stable matching induced by (x, ψ) , and let $s_0 \in \mathcal{S}$ denote the outside option, with $u_{i,s_0} = 0$ and capacity $q_{s_0} = |\mathcal{I}|$. Take $y_{i,s} = D_{i,s}$ for $s \neq s_0$. Constraint (101) follows from (70) after discarding the mass on s_0 , constraint (102) is (71), and (103) is (74); since $u_{i,s_0} = 0$, the objective is unchanged. Hence y is feasible for (100)–(103) and $W_\psi(x) \leq \overline{W}^{\text{TP}}$. Equivalently, (100)–(103) is the sample program (69)–(74) with the access constraints (72) and the stability constraints (73) and (76) deleted, so it is a relaxation of every sample, uniformly in (x, ψ) . Deleting (102) in turn decouples the students and leaves $\sum_i \max_s u_{i,s} = \overline{W}^{\text{FC}}$, giving the second inequality, strictly whenever some program is oversubscribed at its top-choice demand. ■

Because (104) holds pointwise in ψ , it also bounds the sample average, and so $\text{Welfare}_{\max} = \overline{W}^{\text{TP}}$ is a valid domain for η .

Three properties make this substitution attractive. First, (100)–(103) is a transportation problem, so its polytope is integral and the LP optimum is attained by an integral assignment; no rounding argument is needed. Second, it carries one variable per ranked student-program pair — 1.24×10^4 on our instances — and solves in milliseconds, which is negligible beside a single master solve, so the tightening is free. Third, the substitution is weakly dominant: since the reported bound is $\min\{\text{Welfare}_{\max}, \text{cut bound}\}$ and $\overline{W}^{\text{TP}} \leq \overline{W}^{\text{FC}}$, replacing the constant can never worsen the reported bound.

On the Block₂ instances ($|\mathcal{I}| = 3953$ applicants, $|\mathcal{S}| = 123$ programs, $k = 6$ zones), $\overline{W}^{\text{FC}} = 19,311.29$ against $\overline{W}^{\text{TP}} = 18,699.31$ for one centroid configuration and 18,968.60 against 18,370.41 for a second, reductions of 611.98 and 598.19. The improvement actually realized in the reported bound lies between zero and that reduction, according to how far the cuts have already descended: we observe 573.19 and 67.76 respectively, the second case being one in which the cuts had already certified a value close to $\overline{W}^{\text{TP}}$.

We emphasize what this does not address. Neither constant prices the zoning restriction itself: $\overline{W}^{\text{TP}}$ still permits every applicant a globally best available seat,

whereas a feasible zoning confines each applicant to the programs co-zoned with their residence, roughly a $1/k$ share of the district. That unpriced restriction, rather than any weakness of the sample programs, accounts for the bulk of the residual gap, and closing it requires a relaxation that reasons about zone membership — for instance a zone-restricted transportation problem, or an outer approximation written per (applicant, zone) pair rather than per access indicator.

C.2 Congestion-Priced Access

The residual gap identified above is the unpriced cost of confinement, and both routes suggested there — a zone-restricted transportation problem and an outer approximation written per applicant rather than per access indicator — turn out to be the same construction viewed from two sides. This subsection develops it. The result replaces the *constant* $\text{Welfare}_{\max}$ with a *function* of the zoning that is still a valid upper bound on stable-matching welfare, still separable across applicants, and — unlike the dual cuts (52) — admits a finite closed-form outer description requiring no recourse LP at all.

Zone-Restricted Transportation Value: Reinstating the access constraints (72) in the transportation problem of Proposition 4 while still deleting the stability constraints (73) and (76) gives, for each zoning x ,

$$T(x) = \max_y \sum_{i \in \mathcal{I}} \sum_{s \in \mathcal{S}} u_{i,s} y_{i,s} \quad (105)$$

$$\text{s.t.} \quad \sum_{s \in \mathcal{S}} y_{i,s} \leq 1 \quad \forall i \in \mathcal{I} \quad (106)$$

$$\sum_{i \in \mathcal{I}} y_{i,s} \leq q_s \quad \forall s \in \mathcal{S} \quad (107)$$

$$0 \leq y_{i,s} \leq a_{i,s}(x) \quad \forall i \in \mathcal{I}, s \in \mathcal{S}. \quad (108)$$

By the argument of Proposition 4, $W_\psi(x) \leq T(x) \leq \bar{W}^{\text{TP}}$ for every seed ψ , with the second inequality following from $a_{i,s}(x) \leq 1$. The quantity $T(x)$ is the tightest bound obtainable without reasoning about stability, but it is the optimal value of an LP parameterized by x ; used directly it would reintroduce exactly the per-candidate recourse solve that makes (52) expensive, and it would again be summarized by a single dual hyperplane over all $|\mathcal{I}| \times |\mathcal{S}|$ access coordinates. Lagrangian relaxation of the capacity constraints removes both objections at once.

Priced Access Bound: Fix a price vector $\pi \in \mathbb{R}_{\geq 0}^{|\mathcal{S}|}$ and define the *priced value* of program s to applicant i as

$$w_{i,s} = u_{i,s} - \pi_s. \quad (109)$$

Let $F_i \subseteq \mathcal{S}$ collect the programs whose access no zoning can revoke — citywide programs and programs with $l(s) = v(i)$ — and write

$$g_i(a_{i,\cdot}) = \max \left\{ 0, \max_{s \in \mathcal{S}} w_{i,s} a_{i,s} \right\}, \quad \phi_i = \max \left\{ 0, \max_{s \in F_i} w_{i,s} \right\}, \quad (110)$$

so that g_i is the best priced value an applicant can reach under $a_{i,\cdot}$, the outside option being available at value 0, and $\phi_i \leq g_i(a_{i,\cdot})$ is the part of it that is guaranteed. The priced-access bound is

$$\overline{W}^{\text{PA}}(x; \pi) = \sum_{s \in \mathcal{S}} q_s \pi_s + \sum_{i \in \mathcal{I}} g_i(a_{i,\cdot}(x)). \quad (111)$$

Proposition 5 (Validity and Dominance). *For every price vector $\pi \geq 0$, every zoning $x \in \Omega(G, C, k, \delta)$ and every tie-breaking seed $\psi \in \Psi$,*

$$W_\psi(x) \leq T(x) \leq \overline{W}^{\text{PA}}(x; \pi), \quad \text{and} \quad \min_{\pi \geq 0} \overline{W}^{\text{PA}}(x; \pi) = T(x). \quad (112)$$

Moreover, if π^* denotes an optimal dual solution to the capacity constraints (102) of the unrestricted problem, then $\overline{W}^{\text{PA}}(x; \pi^*) \leq \overline{W}^{\text{TP}}$ for every x .

Proof. The first inequality is Proposition 4 with (72) retained. For the second, relax (107) into the objective with multipliers $\pi \geq 0$; weak duality gives, for any such π ,

$$T(x) \leq \sum_s q_s \pi_s + \max \left\{ \sum_{i,s} w_{i,s} y_{i,s} : \sum_s y_{i,s} \leq 1, 0 \leq y_{i,s} \leq a_{i,s}(x) \right\}.$$

The remaining program has no constraint coupling distinct applicants, so it separates into $|\mathcal{I}|$ independent problems. Each is a maximization of a linear function over $\{y_{i,\cdot} \geq 0 : \sum_s y_{i,s} \leq 1, y_{i,s} \leq a_{i,s}\}$, whose optimum places unit mass on a single accessible program of largest priced value when that value is positive, and no mass at all otherwise; its value is therefore exactly $g_i(a_{i,\cdot}(x))$, establishing (112). Since (105)–(108) is a feasible bounded linear program, strong duality applies and the Lagrangian dual attains $T(x)$, giving the stated minimum. Finally, each g_i is nondecreasing in $a_{i,\cdot}$ and $a_{i,s}(x) \leq 1$, so $\overline{W}^{\text{PA}}(x; \pi^*) \leq \overline{W}^{\text{PA}}(\mathbf{1}; \pi^*)$; the latter is the Lagrangian dual objective of the unrestricted transportation problem evaluated at its own optimal multipliers, which by strong duality equals $\overline{W}^{\text{TP}}$. ■

Three consequences are worth separating. Validity holds for *every* non-negative π , so the prices are a free modelling choice rather than a quantity that must be computed correctly; taking $\pi = 0$ recovers the zone-restricted first-choice bound, in which every applicant receives their favourite co-zoned program irrespective of capacity. Dominance means the substitution $\text{Welfare}_{\max} \leftarrow \overline{W}^{\text{PA}}(x; \pi^*)$ can never weaken the bound of (58), exactly as $\overline{W}^{\text{TP}}$ could not weaken $\overline{W}^{\text{FC}}$. And the price of using a single fixed π instead of re-optimizing per candidate is precisely $\overline{W}^{\text{PA}}(x; \pi) - T(x)$, which by the last part of (112) vanishes at the restricted problem's own duals; we quantify it below.

Proposition 6 (Exact Threshold Description). *For each applicant i let*

$$\Theta_i = \{\phi_i\} \cup \{w_{i,s} : s \in \mathcal{S} \setminus F_i, w_{i,s} > \phi_i\}.$$

Then for every $\tau \geq \phi_i$ and every $a_{i,\cdot} \in \{0,1\}^{|\mathcal{S}|}$,

$$g_i(a_{i,\cdot}) \leq \tau + \sum_{\substack{s \in \mathcal{S} \setminus F_i \\ w_{i,s} > \tau}} (w_{i,s} - \tau) a_{i,s}, \quad (113)$$

and the family is exact on the binary cube:

$$g_i(a_{i,\cdot}) = \min_{\tau \in \Theta_i} \left[\tau + \sum_{\substack{s \in \mathcal{S} \setminus F_i \\ w_{i,s} > \tau}} (w_{i,s} - \tau) a_{i,s} \right]. \quad (114)$$

Proof. Write $m = g_i(a_{i,\cdot})$ and let $\tau \geq \phi_i$. If $m \leq \tau$ the right-hand side of (113) is at least $\tau \geq m$. If $m > \tau$ then m is attained by some program s^* with $a_{i,s^*} = 1$ and $w_{i,s^*} = m$; since $m > \tau \geq \phi_i$, the definition of ϕ_i forces $s^* \notin F_i$, so s^* contributes the term $(m - \tau)a_{i,s^*} = m - \tau$ and the right-hand side is at least m . This proves (113), hence that the minimum in (114) is at least m .

For the reverse, choose $\tau = m$ if $m > \phi_i$ and $\tau = \phi_i$ otherwise; in both cases $\tau \in \Theta_i$, because $m > \phi_i$ implies $m = w_{i,s^*}$ for some $s^* \in \mathcal{S} \setminus F_i$ as above. Every accessible program satisfies $w_{i,s} \leq m = \tau$ by maximality of m , so no surviving term has $a_{i,s} = 1$, the sum vanishes, and the bracket equals $\tau = m$. ■

Proposition 6 is the structural payoff. The function g_i is piecewise linear, concave and monotone in $a_{i,\cdot}$, and (114) exhibits it as the minimum of at most one more affine function than the number of programs i ranks with coefficients available in closed form from the sorted priced values — no LP, no dual solution, and no dependence on the seed ψ . Each inequality (113) touches only applicant i 's own access row, which is what makes the family exact rather than merely valid: the cut set describes g_i on the whole cube, whereas the dual cut (52) is a single hyperplane tangent at one point of a $|\mathcal{I}| \times |\mathcal{S}|$ -dimensional domain. The restriction $\tau \geq \phi_i$ is not cosmetic. A threshold below ϕ_i would leave the guaranteed value ϕ_i uncovered by any term of (113), since the programs realizing it lie in F_i and carry no access variable, and the inequality would then cut away welfare the applicant retains under every zoning.

Vertex Aggregation: Because access is geographic, $a_{i,s} \equiv a_{v(i),l(s)}$, so applicants sharing a home vertex share an access row and their inequalities may be summed with no loss — the same exact aggregation invoked for (52), but applied to a family that is already per-applicant rather than to a dual hyperplane. Introduce per-vertex epigraph variables $\theta_v \in \mathbb{R}$ and, for any assignment of thresholds $\tau_i \geq \phi_i$ to the applicants of $\mathcal{I}_v$, define

$$c_v(\tau) = \sum_{i \in \mathcal{I}_v} \tau_i, \quad \Lambda_{v,u}(\tau) = \sum_{i \in \mathcal{I}_v} \sum_{\substack{s \in \mathcal{S}_u \setminus F_i \\ w_{i,s} > \tau_i}} (w_{i,s} - \tau_i). \quad (115)$$

Summing (113) over $\mathcal{I}_v$ gives the vertex cut

$$\theta_v \leq c_v(\tau) + \sum_{u \in \mathcal{S}} \Lambda_{v,u}(\tau) a_{v,u} \quad \forall v \in V, \quad (116)$$

valid for every zoning and every admissible τ . A vertex with no resident applicants has $\mathcal{I}_v = \emptyset$, hence $c_v = 0$ and $\Lambda_{v,\cdot} = 0$; together with $\theta_v \geq 0$ this pins $\theta_v = 0$ exactly, which matters because the master instantiates one epigraph variable per vertex and an uncut variable would otherwise be free up to its domain.

Note the exact structural correspondence with the choice-set cut (39): $c_v(\tau)$ plays the role of the incumbent utility $U_v(S_v^t)$ and $\Lambda_{v,u}(\tau)$ the role of the marginal gains $\Delta_{t,s,v}^+$, while the losses $\Delta_{s,v}^-$ are represented not by a separate term but by the choice of a lower threshold. Consequently the same CP-SAT and MIP master solvers serve all three formulations unchanged.

Master Problem and Separation: The priced-access master problem is

$$\max_{x,a,\theta} \sum_{v \in V} \theta_v \quad (117)$$

$$\text{s.t. } x \in \Omega(G, C, k, \delta) \quad (118)$$

$$\theta_v \leq c_v(\tau^{(r)}) + \sum_{u \in S} \Lambda_{v,u}(\tau^{(r)}) a_{v,u} \quad \forall v \in V, r = 1, \dots, R \quad (119)$$

$$a_{v,u} \iff \bigvee_{z \in [k]} (x_{v,z} \wedge x_{u,z}) \quad \forall (v,u) \in \mathcal{A}^{\text{PA}} \quad (120)$$

$$0 \leq \theta_v \leq \sum_{i \in \mathcal{I}_v} \max\{0, \max_s w_{i,s}\} \quad \forall v \in V, \quad (121)$$

where $\mathcal{A}^{\text{PA}} = \{(v,u) \in V \times S \mid \exists r, \Lambda_{v,u}(\tau^{(r)}) \neq 0\}$ is instantiated on demand exactly as $\mathcal{A}^*$ is in (57). The constant $\sum_s q_s \pi_s$ is added to the objective when its value is read as a welfare bound; being independent of x , it does not affect the maximizer. Given a candidate zoning x^r , separation requires no optimization: for each applicant, sort the accessible priced values and emit the thresholds

$$\tau_i^{(r,\ell)} = \max \left\{ \phi_i, \text{ the } (\ell+1)\text{-th largest } w_{i,s} \text{ over } \{s : a_{i,s}(x^r) = 1\} \right\} \quad (122)$$

for $\ell = 0, \dots, L-1$, taking the outside value 0 once an applicant's accessible options are exhausted. Level $\ell = 0$ is tight at x^r by the exactness argument of Proposition 6; levels $\ell \geq 1$ price what the applicant loses when their ℓ best accessible programs are withdrawn, which is the direction a family generated only at the incumbent cannot see. All L levels are aggregated by (115) into L cuts per vertex per iteration, at a cost of one pass over the preference lists.

Theorem 2 (Finite Convergence to the Bound). *Suppose each master problem (117)–(121) is solved to optimality and duplicate cuts are discarded. Then the priced-access cutting-plane algorithm terminates finitely at a zoning x^* maximizing $\bar{W}^{\text{PA}}(\cdot; \pi)$ over $\Omega(G, C, k, \delta)$, and*

$$\max_{x \in \Omega} \frac{1}{|\Psi|} \sum_{\psi \in \Psi} W_\psi(x) \leq \bar{W}^{\text{PA}}(x^*; \pi). \quad (123)$$

Proof. By Proposition 6 every cut (116) is valid for all $x \in \Omega$, so the master maximizes an outer approximation of $\sum_v \sum_{i \in \mathcal{I}_v} g_i$ and its optimum weakly

exceeds $\max_x \bar{W}^{\text{PA}}(x; \pi) - \sum_s q_s \pi_s$ at every iteration. The threshold set Θ_i is finite and (122) draws from it, so only finitely many distinct cuts (116) exist; discarding duplicates therefore admits finitely many iterations that add a cut. At an iteration adding none, the level-0 cuts at x^r are already present, and by exactness they force $\theta_v = \sum_{i \in \mathcal{I}_v} g_i(a_{i,\cdot}(x^r))$; the master objective thus equals the true value of $\sum_i g_i$ at x^r while dominating it everywhere on Ω , so x^r is a maximizer. Inequality (123) follows because Proposition 5 holds pointwise in ψ , hence for the sample average, and x^* maximizes the dominating function. ■

Two limitations deserve emphasis, since (123) is a statement about the bound and not about the matching. First, $\bar{W}^{\text{PA}}$ strictly exceeds sample-average stable-matching welfare, so its maximizer need not maximize welfare; the algorithm returns a certified optimum of a surrogate, and the usefulness of that optimum rests on the surrogate’s bias being close to constant across the zonings the master visits. Second, Theorem 2 presumes exact master solves, whereas a time-limited CP-SAT solve returns an incumbent; under a per-iteration limit the method retains validity of (123) for the cuts generated but forfeits the optimality claim, and terminates on the iteration budget rather than on the absence of separation.

Decomposition of the Bias: Proposition 5 splits the residual optimism into two independently interpretable pieces,

$$\underbrace{\bar{W}^{\text{PA}}(x; \pi^*) - T(x)}_{\text{fixed rather than re-optimized prices}} + \underbrace{T(x) - \frac{1}{|\Psi|} \sum_{\psi} W_{\psi}(x)}_{\text{stability discarded}}. \quad (124)$$

On the Block₂ instance with $k = 6$ zones and $|\Psi| = 15$ seeds, evaluated at the final zonings of five different strategies and with π^* taken from the single unrestricted transportation LP of Proposition 4, which gives $\bar{W}^{\text{TP}} = 18,699.31$ for this centroid configuration. Writing $\bar{W}_{\Psi}(x) = \frac{1}{|\Psi|} \sum_{\psi} W_{\psi}(x)$ for sample-average stable-matching welfare:

zoning from	$\bar{W}^{\text{PA}}$	$T(x)$	$\bar{W}_{\Psi}(x)$	prices	stability
SAA	15,084.29	14,997.76	14,822.77	86.53	175.00
short bursts	14,735.40	14,647.44	14,486.66	87.96	160.78
choice set	14,635.02	14,486.79	14,324.86	148.23	161.94
MID decomposition	14,220.26	14,098.75	13,884.10	121.50	214.65
MID	14,203.24	14,011.49	13,849.74	191.75	161.75

The last two columns are the two terms of (124). The zoning-independent constant $\bar{W}^{\text{TP}}$ overstates sample-average welfare by 3,877 to 4,850 depending on the zoning evaluated; $\bar{W}^{\text{PA}}$ overstates it by 249 to 354, of which 87 to 192 is attributable to holding π fixed and 161 to 215 to discarding stability. Two features of the table matter more than its absolute magnitudes. The bound is valid at every one of the five zonings and at every individual seed, not merely on average, as Proposition 5 requires. And the overstatement is nearly constant across zonings produced by five structurally unrelated methods and spanning

almost a thousand units of welfare, which is the empirical condition under which maximizing $\overline{W}^{\text{PA}}$ approximates maximizing welfare — a regularity we observe rather than a property we can prove, and the reason the limitations noted after Theorem 2 are stated as limitations rather than dismissed.

Role of the Prices: Setting $\pi = 0$ isolates what congestion pricing contributes. The resulting surrogate — each applicant’s favourite co-zoned program, capacity ignored — is cheaper to optimize and in our runs converges faster, terminating on the absence of separation with a model-versus-realized gap of 0.01%; but it converges to a materially worse zoning, because an unpriced surrogate is indifferent between a program with spare seats and an oversubscribed one and is therefore rewarded for concentrating demand on popular schools. Charging π^* recovered 1,357 units of realized welfare relative to $\pi = 0$ on one instance at a matched iteration allowance — a single-instance observation, but a difference an order of magnitude larger than the bias figures tabulated above. This is the clearest illustration of the gap between Theorem 2 and the objective one actually cares about: the algorithm certifies optimality for whichever surrogate it is given, and the surrogate must be the right one.

D Dantzig–Wolfe Decomposition

The MID objective admits a decomposition whose columns are entire feasible zones when no program offers citywide access. The restriction is substantive: a citywide program couples applicants in different zones through a shared capacity constraint, preventing zone welfare from being evaluated independently. In this appendix we remove citywide schools and their programs from the market and from the school and capacity counts. Applicants remain at their original residential vertices; their preference lists retain the order, utilities and priority tiers of the remaining programs. An applicant with no remaining acceptable program receives the outside option, of utility zero.

We develop the decomposition for the finite-grid MID model at a fixed geographic resolution. Fix an integer lottery scale $L \geq 1$ and restrict each cutoff to $P_s = p_s/L$, where p_s is a nonnegative integer. Consequently, all assignment probabilities are multiples of $1/L$. The result below certifies the optimum of this specified finite-grid model; no approximation claim about the continuous-cutoff optimum is needed.

D.1 Feasible Zones and Their Welfare

For each label $z \in [k]$, let $\mathcal{F}_z$ be the family of all admissible individual zones $A \subseteq V$. Membership requires $A \neq \emptyset$, connectedness of $G[A]$, and the capacity, school-count and demographic inequalities for that zone. These inequalities are

local to A : for example,

$$\underline{s} \leq \sum_{v \in A} s(v) \leq \bar{s}, \quad (125)$$

$$(1 - \delta^-)N(A) \leq C(A) \leq (1 + \delta^+)N(A), \quad (126)$$

$$(F - \delta_F)N(A) \leq \sum_{v \in A} f_v \leq (F + \delta_F)N(A), \quad (127)$$

$$(R_e - \delta_R)N(A) \leq \sum_{v \in A} r_{v,e} \leq (R_e + \delta_R)N(A) \quad \forall e \in \mathcal{E}. \quad (128)$$

Here $\underline{s}$ and $\bar{s}$ are the prescribed school-count bounds, such as those in (6); either aggregate capacity bound may be omitted if it is not imposed. Explicit candidate labels and fixed assignments also enter $\mathcal{F}_z$. If anchors are required, include $c_z \in A$ and exclude every other centroid from A . For an unanchored formulation, omit these restrictions. The same decomposition applies to either choice, provided seeding, pricing and the master use the same families $\mathcal{F}_z$. Individual admissibility does not require that $V \setminus A$ itself be an admissible zone.

Let $\mathcal{G}_A = \{g \in \mathcal{G} : v(g) \in A\}$ and $\mathcal{S}_A = \{s \in \mathcal{S} : l(s) \in A\}$. Define $W_L(A)$ as the greatest MID welfare obtainable by these applicants using these programs at lottery scale L . To express it on an integer mass scale, write $\hat{R}_{g,r} = LR_{g,r}$ and $\hat{d}_{g,r} = Ld_{g,r}$. Given access indicators $a_{g,s}$, the complete recurrence is

$$\hat{t}_{s,\rho} = \min\{L, \max\{p_s - L\rho, 0\}\}, \quad (129)$$

$$\hat{e}_{g,r} = a_{g,s(g,r)}\hat{t}_{s(g,r),\rho(g,r)} + (1 - a_{g,s(g,r)})L, \quad (130)$$

$$\hat{R}_{g,0} = L, \quad \hat{R}_{g,r} = \min\{\hat{R}_{g,r-1}, \hat{e}_{g,r}\}, \quad (131)$$

$$\hat{d}_{g,r} = \hat{R}_{g,r-1} - \hat{R}_{g,r}. \quad (132)$$

For a fixed A , set $a_{g,s} = 1$ exactly when $g \in \mathcal{G}_A$ and $s \in \mathcal{S}_A$. Using all types and ranks in (129)–(132) then gives zero assigned mass to every omitted applicant. The zone value is

$$\begin{aligned} W_L(A) = \max_{p, \hat{R}, \hat{d}} \quad & \frac{1}{L} \sum_{g \in \mathcal{G}} n_g \sum_{r=1}^{R_g} u(g, r) \hat{d}_{g,r} \\ \text{s.t.} \quad & (129)–(132), \\ & \sum_{g \in \mathcal{G}} \sum_{r: s(g,r)=s} n_g \hat{d}_{g,r} \leq Lq_s \quad \forall s \in \mathcal{S}, \\ & p_s \in \{0, \dots, \bar{p}_s\} \quad \forall s \in \mathcal{S}, \end{aligned} \quad (133)$$

where $\bar{p}_s = L(1 + \max_{g,r: s(g,r)=s} \rho(g, r))$, with an empty maximum taken as zero. This bound permits rejection of every applicant, so (133) is always feasible. All remaining masses are integers between zero and L .

Proposition 7 (Least-Cutoff Zone Value). *For every admissible zone A , the value $W_L(A)$ equals the welfare at its componentwise least capacity-feasible cutoff vector.*

Proof. Start with $p = 0$. Whenever a program is overloaded, raise only its cutoff to the smallest integer value that clears its capacity, holding the other cutoffs fixed. Demand is nonincreasing in a program's own cutoff, and $\bar{p}_s$ clears it. Each update strictly increases a bounded integer coordinate, so the procedure terminates at a capacity-feasible vector p^* .

Let $\tilde{p}$ be any capacity-feasible vector. Initially $p \leq \tilde{p}$. If this inequality holds before an update of program s , then evaluating its demand at $p_s = \tilde{p}_s$ while keeping all other cutoffs at their current, weakly smaller values yields demand no greater than at $\tilde{p}$. Indeed, lower cutoffs at earlier choices can only reduce the mass remaining to demand s ; later choices do not affect its demand. Thus the smallest clearing update satisfies $p_s \leq \tilde{p}_s$. Induction gives $p^* \leq \tilde{p}$, proving leastness.

Set $u(g, R_g + 1) = 0$. By (131), lower cutoffs weakly decrease every remaining mass. For a nonempty preference list, the type's welfare is

$$\frac{n_g}{L} \sum_{r=1}^{R_g} u(g, r) \hat{d}_{g,r} = n_g u(g, 1) - \frac{n_g}{L} \sum_{r=1}^{R_g} (u(g, r) - u(g, r+1)) \hat{R}_{g,r}. \quad (134)$$

Every coefficient of a remaining mass is nonpositive because utilities are ordered. Hence p^* weakly maximizes welfare among all capacity-feasible cutoff vectors. Empty lists contribute zero. ■

D.2 The Zone-Column Master

For $A \subseteq V$, define its half-perimeter by

$$b(A) = \frac{1}{2} \sum_{\substack{\{u,v\} \in E \\ |\{u,v\} \cap A| = 1}} w_{u,v}. \quad (135)$$

With $w_{u,v} = 1$ this measures half the number of edges leaving A . Let $\bar{b} \geq 0$ be an optional district boundary budget. Introduce a binary variable $\lambda_{z,A}$ selecting zone A for label z . The master is

$$\max_{\lambda} \sum_{z \in [k]} \sum_{A \in \mathcal{F}_z} W_L(A) \lambda_{z,A} \quad (136)$$

$$\text{s.t.} \quad \sum_{z \in [k]} \sum_{\substack{A \in \mathcal{F}_z \\ v \in A}} \lambda_{z,A} = 1 \quad \forall v \in V, \quad (137)$$

$$\sum_{A \in \mathcal{F}_z} \lambda_{z,A} = 1 \quad \forall z \in [k], \quad (138)$$

$$\sum_{z \in [k]} \sum_{A \in \mathcal{F}_z} b(A) \lambda_{z,A} \leq \bar{b}, \quad (139)$$

$$\lambda_{z,A} \in \{0, 1\} \quad \forall z \in [k], A \in \mathcal{F}_z. \quad (140)$$

If compactness is unrestricted, delete (139).

Proposition 8 (Exact Decomposition). *The integer master (136)–(140) has the same optimal value as finite-grid MID zoning over the prescribed families $\mathcal{F}_z$.*

Proof. Constraints (137) and (138) select exactly one nonempty admissible zone per label and cover each vertex exactly once. Thus they define precisely the admissible partitions. For any such partition $\mathcal{P} = (A_1, \dots, A_k)$, the sets $\mathcal{G}_{A_z}$ partition the applicants and the sets $\mathcal{S}_{A_z}$ partition the programs. No applicant can use a program in another zone, so both the MID recurrence and the program capacity constraints separate across zones. Independent optimizers of (133) therefore combine into a feasible district allocation, and every district allocation restricts to a feasible allocation in each zone. Consequently,

$$W_L^{\text{MID}}(\mathcal{P}) = \sum_{z \in [k]} W_L(A_z). \quad (141)$$

Finally, each cut edge occurs in exactly two zone perimeters, so $\sum_z b(A_z)$ is precisely the district boundary cost. Both objective and feasibility are preserved. ■

D.3 Exact Pricing of an Individual Zone

Maintain a restricted pool $\mathcal{F}'_z \subseteq \mathcal{F}_z$ and relax (140) to $\lambda \geq 0$. Let α_v and γ_z be the unrestricted dual multipliers of (137) and (138), and let $\mu \geq 0$ be the multiplier of (139). The complete dual requires

$$\sum_{v \in A} \alpha_v + \gamma_z + \mu b(A) \geq W_L(A) \quad \forall z \in [k], A \in \mathcal{F}_z. \quad (142)$$

Its objective is $\sum_v \alpha_v + \sum_z \gamma_z + \mu \bar{b}$. Take $\mu = 0$ and omit the budget term when there is no boundary constraint. Pricing for label z maximizes the reduced welfare

$$\max_{A \in \mathcal{F}'_z} \left\{ W_L(A) - \sum_{v \in A} \alpha_v - \gamma_z - \mu b(A) \right\}. \quad (143)$$

A strictly positive optimum identifies an improving column; nonpositive optima for all labels certify that the restricted LP solves the full LP relaxation.

To solve (143) globally, introduce binary memberships $y_v = \mathbf{1}\{v \in A\}$ and optimize them jointly with the cutoff and mass variables in (133). Access becomes

$$a_{g,s} = y_{v(g)} y_{l(s)}, \quad a_{g,s} \leq y_{v(g)}, \quad a_{g,s} \leq y_{l(s)}, \quad a_{g,s} \geq y_{v(g)} + y_{l(s)} - 1. \quad (144)$$

All access variables are binary. If $v(g) = l(s)$, access equals $y_{v(g)}$, so omitted vertices contribute no applicants.

Connectedness has an exact single-commodity flow description. Replace each undirected edge by both arcs, let $m = |V|$, choose a root with binary variables

ξ_v , and let $\tau_v \geq 0$ be the supply injected at vertex v . Nonnegative arc flows φ_{uv} satisfy

$$\sum_{v \in V} \xi_v = 1, \quad \xi_v \leq y_v \quad \forall v \in V, \quad (145)$$

$$0 \leq \tau_v \leq m\xi_v \quad \forall v \in V, \quad (146)$$

$$\tau_v + \sum_{u \in N(v)} (\varphi_{uv} - \varphi_{vu}) = y_v \quad \forall v \in V, \quad (147)$$

$$0 \leq \varphi_{uv} \leq my_u, \quad \varphi_{uv} \leq my_v \quad \forall (u, v). \quad (148)$$

A selected component without the root would have positive total demand and zero net supply, contradicting (147). Conversely, a connected selected set can route one unit to every selected vertex along a rooted spanning tree. Thus the formulation enforces exactly nonempty connectedness. Express (125)–(128) by replacing each sum over A with its corresponding membership-weighted sum. Candidate labels and fixed assignments set the appropriate y_v to zero or one.

For each edge, a binary variable $\beta_{uv} = |y_u - y_v|$ is imposed by

$$\begin{aligned} \beta_{uv} &\geq y_u - y_v, & \beta_{uv} &\geq y_v - y_u, \\ \beta_{uv} &\leq y_u + y_v, & \beta_{uv} &\leq 2 - y_u - y_v. \end{aligned} \quad (149)$$

The pricing objective is therefore

$$\frac{1}{L} \sum_g n_g \sum_{r=1}^{R_g} u(g, r) \hat{d}_{g,r} - \sum_v \alpha_v y_v - \gamma_z - \frac{\mu}{2} \sum_{\{u,v\} \in E} w_{u,v} \beta_{uv}. \quad (150)$$

Together with (129)–(132), program capacities and the membership constraints above, this is an exact pricing formulation by Proposition 7. It does not require any simultaneous partition of the complement.

For completeness, each nonlinear expression has a bounded binary linearization. If $a, b \in [0, M]$, impose $h = \min(a, b)$ using a binary χ and

$$0 \leq h, \quad h \leq a, \quad h \leq b, \quad h \geq a - M\chi, \quad h \geq b - M(1 - \chi). \quad (151)$$

If $a \in [-M, M]$, impose $h = \max(a, 0)$ using

$$h \geq a, \quad h \geq 0, \quad h \leq a + M(1 - \chi), \quad h \leq M\chi. \quad (152)$$

Use $M = \bar{p}_s$ for the threshold operations and $M = L$ for the remaining-mass recurrence. Finally, (130) is equivalent to

$$\begin{aligned} 0 &\leq \hat{e}_{g,r} \leq L, & \hat{e}_{g,r} &\geq \hat{t}_{s(g,r), \rho(g,r)}, \\ \hat{e}_{g,r} &\geq L(1 - a_{g,s(g,r)}), & \hat{e}_{g,r} &\leq \hat{t}_{s(g,r), \rho(g,r)} + L(1 - a_{g,s(g,r)}). \end{aligned} \quad (153)$$

Cutoff, threshold and mass variables are integer; flow variables may be continuous. These formulations retain every rank of every preference list.

Proposition 9 (Pricing-Corrected Upper Bound). *Suppose $\bar{r}_z$ bounds the maximum reduced welfare in (143) from above for each label. Then*

$$\text{UB} = \sum_v \alpha_v + \sum_z \gamma_z + \mu \bar{b} + \sum_z \max\{0, \bar{r}_z\} \quad (154)$$

is an upper bound on the complete LP and hence on every admissible partition at that branch.

Proof. Set $\gamma'_z = \gamma_z + \max\{0, \bar{r}_z\}$. For every $A \in \mathcal{F}_z$, the reduced-cost bound implies

$$\sum_{v \in A} \alpha_v + \gamma'_z + \mu b(A) \geq W_L(A).$$

Thus (α, γ', μ) is feasible for the complete dual, and its objective is (154). Weak duality proves the claim. ■

The bound remains valid before column generation has converged, provided every $\bar{r}_z$ is a global upper bound on its pricing problem. In particular, finding no improving zone by a local search supplies no such certificate.

D.4 Branching and Finite Convergence

Exact column generation solves the master LP, whose optimum may still be fractional. Define the geographic assignment marginals

$$\bar{x}_{v,z} = \sum_{\substack{A \in \mathcal{F}'_z \\ v \in A}} \lambda_{z,A}. \quad (155)$$

If $0 < \bar{x}_{v,z} < 1$, branch into $x_{v,z} = 0$ and $x_{v,z} = 1$. In the first child, exclude v from every column of label z . In the second, require v in every column of label z and exclude it from every other label. Apply each decision both to the stored pool and to all future pricing problems. These operations define branch-specific subfamilies of $\mathcal{F}_z$ without adding any cross-zone constraint to the pricing problem.

ReCom supplies an optional initialization: sample feasible partitions, retain their constituent zones, and evaluate each distinct labelled membership once. Accepted partitions need not improve the current welfare to contribute useful columns. The column pool can combine zones from different samples. A finite amount of sampling precedes the exact search; neither coverage of the feasible partition space nor a mixing assumption is required for correctness.

Theorem 3 (Finite Convergence to Optimal MID Welfare). *Fix a finite graph, a finite lottery scale L , and the zone families $\mathcal{F}_z$. Suppose all restricted LPs and pricing problems are solved globally in exact arithmetic, and generated columns are retained and deduplicated by label and membership. Then Algorithm 2 terminates after finitely many steps at a globally optimal finite-grid MID partition, or proves that no feasible partition exists.*

Proof. Each family contains at most $2^{|V|} - 1$ memberships. At an optimal restricted LP solution, a stored column has nonpositive reduced cost. Therefore every strictly positive priced column is new, and each column-generation

Algorithm 2 Zone-Column Branch-and-Price for Finite-Grid MID

Require: Graph G , families $\mathcal{F}_z$, lottery scale L , optional budget $\bar{b}$.

- 1: Seed pools with finitely many feasible ReCom samples, or leave empty.
 - 2: Retain the best sampled partition, if any; initialize the root search node.
 - 3: **while** an unexplored node remains **do**
 - 4: Select a node; restrict pools and pricing to its branch decisions.
 - 5: **if** the restricted master is infeasible and the branch's columns are exhausted **then**
 - 6: Discard the node and continue.
 - 7: **end if**
 - 8: Solve the welfare LP; globally price every label using (143).
 - 9: Add positive-reduced-cost columns and repeat until none remain.
 - 10: Optionally improve the incumbent using an integer restricted master.
 - 11: **if** the certified node bound does not exceed incumbent welfare **then**
 - 12: Discard the node.
 - 13: **else if** all marginals (155) are integral **then**
 - 14: Recover the partition, update the incumbent, and discard the node.
 - 15: **else**
 - 16: Branch on a fractional marginal and retain both children.
 - 17: **end if**
 - 18: **end while**
 - 19: **return** the incumbent, or infeasibility if no partition was found.
-

loop can add only finitely many columns. When all pricing optima are nonpositive, the restricted dual is feasible for the complete branch-specific master, and strong duality certifies its LP optimum. A node whose restricted master is still infeasible once its column family is exhausted admits no feasible partition, every admissible zone of that branch having been generated by then. The pricing models themselves are bounded mixed-integer programs: cutoff and membership variables have finite domains, and the flow variables have finite bounds.

Suppose all marginals (155) are integral. For each label, nonnegative column weights sum to one. A marginal of one forces every positive-weight column to contain that vertex; a marginal of zero excludes it from all such columns. Their memberships therefore coincide. Deduplication leaves one column of weight one per label, yielding an integer partition.

Otherwise, branching on a fractional marginal is disjoint and exhaustive for integer partitions and fixes a previously unfixed binary assignment. Pricing respects all these decisions. Tree depth is therefore at most $k|V|$, and the tree is finite. Infeasibility pruning removes no feasible partition, while Proposition 9 justifies bound pruning. An integer restricted master can improve the incumbent but is never needed to justify a bound. On exhaustion of the tree, no feasible partition has welfare greater than the incumbent; Proposition 8 identifies its value with the global finite-grid MID optimum. If no incumbent exists, every branch was infeasible. ■

The guarantee holds for the fixed graph and lottery scale, without a polynomial bound on computation. If stopped early, incumbent welfare is a lower bound; its maximum with the valid bounds of unresolved nodes is a global upper bound. Closing this gap certifies optimality. ReCom contributes candidate zones, while exact pricing and branching establish the guarantee.